

BLOCK - 4
STRATEGIC ENABLERS

BLOCK 4 STRATEGIC ENABLERS

This block discusses some very important tools or strategic enablers that help a firm implement the strategies much more effectively. It explains the importance of Information Technology (IT) to a firm in today's competitive environment and the various factors that have to be considered while designing a firm's IT architecture and infrastructure and the critical role of IT in fostering innovation and enhancing the performance of the firm. It deals with methods for identification and selection of R&D projects and the steps involved in developing a R&D strategy. It focuses on the concept of Knowledge Management (KM), sources and types of knowledge, benefits of KM and challenges for implementing KM in an organisation. Finally it deals with another important strategic enabler, i.e., innovation and the various factors influencing innovation and creativity. This block consists of the following **four units**:

Unit – 9: *IT* and Strategy

Unit – 10: Technology and R&D

Unit – 11: Knowledge Management (KM)

Unit – 12: Innovation

UNIT 9 *IT* AND STRATEGY

Objectives

After reading this unit, you should be able to:

- explain the importance of IT in strategy;
- identify the various factors, which influence the choice of IT architecture and infrastructure;
- discuss the role of IT innovation and performance of a firm;
- describe e-business and steps in e-business plan; and
- understand the role of IT in service quality.

Structure

- 9.1 Introduction
- 9.2 IT and Strategy
- 9.3 Use of IT in Strategy Implementation
- 9.4 IT for Innovation and Performance
- 9.5 E-Business
- 9.6 IT in Service Sector
- 9.7 Summary
- 9.8 Key Words
- 9.9 Self-Assessment Questions
- 9.10 References /Further Readings

9.1 INTRODUCTION

Increasing competition, higher performance levels, globalization, and liberalization are examples of the immense changes that most of the organizations face today. Companies are forced to continuously and organically re-organize and re-shape themselves, meanwhile changing functional hierarchies into flexible, high performance network organizations. To cope with these challenges, organizations need to consider information technology (IT) as an important factor; not only to strengthen operational efficiency and effectiveness, but also to quickly and constantly respond to customer needs and competitive pressures with IT-enabled products, services, and distribution channels, and IT-enabled links with customers, suppliers, and other stakeholders.

The rapid growth of technological innovations and the synthesis of information technology and computer networks are radically changing the way companies compete. Many business enterprises are making strategic commitments to technology for the purpose of gaining and sustaining competitive advantage in their industry. The creation of a competitive advantage through the use of information technology (IT) requires business executives to control this vital corporate resource and manage its use.

Over the last three/four decades, information technology has progressed from a purely academic topic to the point where it has been absorbed into the mainstream. There is hardly a company of any size that does not depend on information technology for its operational success. In spite of this success, information technology is often regarded as necessary evil consuming vast amount of resources yet having little strategic impact on an organization. Information technology is transactional and operational; it aids an organization to automate many of the main operational processes; it enhances efficiency but its effectiveness is often a suspect.

Organizations see information technology as contributing factor to some of their goals - but they tend to be those associated with financial performance rather than with performance on the key and core strategic aspects. The nature and role of information technology has developed over the years. The original notion and practice involved the automation of simple, and single, existing manual and pre-computer mechanical processes. The next stage saw information technology deployed to achieve integration and rationalization of these separate, single systems. In each of these approaches, IT was (and largely still is) used primarily as an operational support tool.

By contrast with other waves of new technology, IT has a number of distinctive features that make its potential to influence social change very significantly. These features include:

- **Ubiquitous Application:** Users irrespective of the type of business or role they perform can apply information technology in many different ways. An e-mail system, access to the Internet and data processing capability are just as relevant for a hospital as for a component manufacturer. In fact it is highly likely that they use similar hardware and software and could communicate and exchange data quickly and easily should they need to.
- **Dramatic Rate of Cost Decline:** The price of processing power, data storage and transmission has fallen drastically. Today a simple electronic toy contains more processing power than was used on the Apollo space programme.
- **Universal Ownership:** The increasing utility and ever lower cost of hardware and software means that they are now almost universally adopted. However the availability of bandwidth to enable rapid communication and transmission of data remains problem in India and is, therefore, a block to further development.
- **Exponential Growth:** Continuous, rapid development and innovation means that the trends to cost reduction and capacity increase continue. The earliest telegraph equipment, using movable arms, had a capacity of 0.2 bits per second while a fibre optic cable has a capacity of more than 10 billion bits per second. These developments suggest that the pace of change is going to be at least maintained and almost certainly increase due to endogenous growth.

A review of the literature suggests that many factors independently and collectively influence a firm's competitiveness (Porter, 1980). However, a growing stream of research since 1980 has examined the concept of

IT as a powerful competitive factor for organizations (Porter and Miller, 1985; Barney, 1999). Recent research has suggested the need for a more integrative approach between IT objectives and business strategy. Other research has examined the value of IT as a viable competitive factor resulting in increased productivity, improved profitability, and value for customers. Studies on the role of IT in competitiveness have been primarily focused on large organizations. Few studies have emphasized the strategic importance and the value of IT in competitiveness. However, in today's global market, and with the use of Internet and electronic business, even small and medium-size enterprises (SMEs) employ IT to increase their competitive position along with their large counterparts. This is believed to be due to fewer obstacles associated with systems integration and more flexibility to implement change.

In order to take full advantage of IT and to compete in the global business environment, the top management must recognize the strategic value of IT and exploit it. However, very few understand technology issues to incorporate them into their strategic plans. For this reason, IT professionals must identify information needs of the organization and develop an IT strategy that is in line with the overall corporate strategic plan. User departments and top management should participate in the development of an IT plan and communicate their needs to IT professionals. An IT plan includes factors such as: a computer hardware and software requirement, systems definition, changes to the existing systems and procedures, and the schedules and resource requirements for each project.

9.2 IT AND STRATEGY

Information technology (IT) in every organization normally evolves from a means to improve the efficiency and effectiveness of an organization to a means to influence the strategic position of the company. The way in which management controls IT has changed simultaneously. In the first stage of IT implementation, efficiency is the primary goal and the attention of management is mainly focused on technology. In this stage, the IT professional is generally an outside consultant, who decides what is best for the organization. In subsequent stages, the effective functioning of the organization becomes as important a goal as efficiency. The management then becomes conscious of the fact that, next to technology, the design and structure of the organization is a decisive factor. User participation, information planning and the appointment of steering committees are indications of this. These organizations increasingly recognize the need for a methodical approach to IT planning, as a result of disruptions in management, reorganization, cost increase, or new usage possibilities.

To a large extent, organizations which have more experience with automation realize that IT can, not only improve the efficiency and effectiveness, but also that it is of decisive importance to the company's success. IT planning, subsequently, acquires a strategic quality in these organizations and, in fact, functions as a catalyst in all this. These organizations set up business architecture and IT architecture, based on an objective, qualitative and quantitative analysis into the current use of information technology.

A good strategy cannot easily be copied by competitors, because of the organizational, financial, social and technical cost and the trial period involved in attaining the strategy. Every organization has its own profile, environment and aims. Strategy indicates how the organizational structure should be designed and what the use of IT should be and should, therefore, be sufficiently concrete and specific. A specific strategy leads to a unique interpretation of the architecture and the infrastructure.

A good strategy focuses less on the product or the service itself and more on the delivery of services, reputation, etc. This is especially important for products that have been “commoditized”. That is the reason why strategies differ in the same sector to a high degree. For example, difference in strategy arises when the focus of a company is either the top or the bottom of the market. For instance, quality, product features and product innovation are applied in a very different ways in different segments of the same market. This leads to very different information needs and to a unique use of IT within the same sector. For example, quality, customization and product innovation are of vital importance to a premium car such as Mercedes while standardization, quality and low cost/price are vital to a budget car such as Maruti 800. In both cases, IT should strongly support these business processes and constantly provide the management with appropriate information.

Steps in Designing IT Architecture and Infrastructure

In practice the realization of strategy, architecture and infrastructure is an interactive process. The development of an IT architecture starts with the business strategy. The business strategy will determine the “business architecture” – the organization structure and the organization processes and the business architecture in turn will determine the “IT architecture” and “IT infrastructure”.

Business Vision

The business strategy should be precise and unambiguous for proper design and implementation of IT architecture. A good strategy starts with a clear business vision and it clearly spells out the direction the company is expected to follow in the years to come. A good business vision should be “inspiring enough to cause people to consider that it is worthwhile to give it their time and energy”. A business vision should be a challenge: not vague, but specifically focused on the organization. This is vital since it provides a framework for strategy formulation.

Business Objectives

A business objective can be defined as the choice of policy that the company wishes to pursue to realize the business vision. The choices may concern the well-known “Ps” of marketing, namely, product, price, promotion and place and also strategic factors such as competition, clients, suppliers and replacement products. For example, the business objectives may deal with policy alternatives such as, market share versus profitability; short-term versus long-term orientation or; growth versus consolidation. Though the objectives provide overall direction for the organizations, they still provide specific direction for IT architecture and infrastructure.

The business vision and the objectives by themselves do not sufficiently explain which aspects of the business should be reinforced, or what competencies the company should possess to perform well. Competencies can be of an economical, technological and an organizational nature, as well as of a social nature. By identifying the required competencies and establishing a set of performance measures, a firm can now translate the business vision into a number of concrete items of which it should be capable. The existing organization and IT will have to be modeled on this basis. Performance measures specify the capabilities, which should be developed. The design criteria are the essential attributes required for success on the basis of which the organization and the IT are modeled. Performance measures can be very simple statements such as; “An important client should have a contact point within the bank and, therefore, we should appoint a key account manager.” In practice this means a total transformation and it is quite natural that the senior management must guide such reorganization.

Business Architecture

Strategy has important organizational and technological consequences. Therefore, formulating a business strategy is insufficient to make the organization perform. The first step, after developing the business strategy is therefore to determine the business architecture required to operate business processes. The effects of the improved use of IT can only be expected when the company processes have also been effectively structured and when the responsibilities are well defined. This has consequences for the organization structure. In the business architecture, attention has to be, therefore, paid to the redesign of the organization and the way in which the organization should function in the future.

The business architecture is divided into three closely related blueprints. These three together constitute the business architecture:

Business Function Blueprint: This blueprint describes the dividing of company processes into responsibility fields and how they are put into practice (centralized-decentralized).

Data Access Blueprint: In order to implement the company processes, all information is necessary. That is why the information needs are described.

Application Access Blueprint: In order to approach the information, applications are necessary. In the application access blueprint the necessary applications are described.

IT Architecture

The distinction between business architecture and IT architecture is of major importance. In many organizations the architecture is mainly determined by technical and economical considerations. The organizational aspects are, therefore, mainly realized by means of the technical opportunities (technology push) and not on the basis of strategic and/or organizational considerations. Within the scope of the business needs, the business architecture offers the possibility to choose the best IT solutions. In this way, the IT architecture fulfils a “bridge” function between, on the one hand, business demands on information supply and, on the other, technological opportunities.

In the business architecture, the way in which the organization should function in the future is indicated, along with which information needs are important in this. The business architecture describes this in a more functional sense. The IT architecture translates the more functional description into technical solutions. The difference between business architecture and IT architecture is, in fact, comparable with the distinction between functional and technical design, which has been used by system development methodologies for years.

The business architecture is dominated by managerial and organizational questions (what). The answers to these questions are translated, in the IT architecture, into automation directions (how) and eventually in the choice of specific makes and types of technical means: the IT infrastructure (with what).

IT Infrastructure

The IT infrastructure is described as the setting-up and management of the whole hardware, software and data communication supply, in such a way that the business architecture and IT architecture can be implemented successfully. The IT infrastructure distinguishes itself from the IT architecture in that the concrete services and products are specified. At first it was sufficient to indicate that there was a need for mini-computers and workstations. In this phase it is determined which specific requirements the systems should meet, in terms of speed, distributed databases and the corresponding machines, local/long-distance communication opportunities, co-operative processing and user interfaces. Also, specific brands and types of products are determined.

Components of IT Infrastructure

The IT infrastructure is not only a matter of hardware, but also a complete integration of the information technology supply. Within this scope, the following components are recognized in the IT infrastructure:

IT Components: The components are formed by the various hardware components, consisting of computers, displays, personal computers, printers, data communication connections and disk units. The software for the steering of all these components is included. In the choice of hardware components other factors prevail, such as compatibility (to what extent can computer systems of different makes and models actually exchange data), data communication possibilities, speed of processing, the price/output-relation and the ease of operation.

IT Services: IT services are services, which execute specific assignments for applications. In the past, application programmers had to set up and develop their file organization on their own. Nowadays, the database management systems take over a large part of these activities. The same applies to the network management. The software involved takes care of the accurate operation and control of the data communication facilities and offers many facilities for management.

IT Control Instruments: The previous components, services and facilities cannot be put in, in a sufficiently effective and efficient way, if attention is not paid to control instruments as well. These instruments consist of:

procedures, methods, techniques and tools for system development and the quality and experience of the automation personnel.

The IT infrastructure will function better if the products and services are connected and are in tune with each other. The company will profit more, and more directly by this. The more one is capable of putting in the IT infrastructure, the better one can do in the primary business processes. The IT infrastructure should not only support the most obvious business processes, it is precisely the support of strategically important business processes, which can lead to the greatest success. Therefore, it is important to select the components and services in such a way that there is sufficient material for a fast and effective creation of new company facilities.

Factors in the IT infrastructure

In developing the IT infrastructure, other factors play a role as well. These factors are, among other things:

User-Friendliness: An example of user-friendliness is the requirement that for every use at every place of work, the same meaning should apply to function-tests. This can also go in the direction of graphic presentations of data, the use of a mouse and the use of pull-up and down menus for the entering of data coding. User-friendliness requires computer capacity and may have consequences for the filling-in of blueprints.

Cost Control: Cost control depends on the question as to whether intelligent terminals are being used, or not, at the workstation, on the installation of applications and the data storage at the workstation. The costs of hardware, communication and control should be weighed against each other.

Hardware Policy: If different computer suppliers are employed, it should be questioned whether it is sufficient to deal with just one specific supplier, because of the desired functionality and the choices made in the IT architecture. Products of different suppliers, however, cannot always be linked.

Safeguarding: When information becomes more and more valuable, it becomes necessary to describe the requirements for the logical and concrete access to, and the use of, applications.

All the factors mentioned influence the parts of the IT infrastructure, and these are decisive in the IT policy. To sum up, the importance of establishing the IT infrastructure is that the IT architecture is defined, in a concrete sense, in the shape of:

- **Descriptions of the necessary computer systems:** From mainframes to personal computers and workstations;
- **Descriptions of the local and central database management systems to be used:** Relational, distributed, data directory/dictionary;
- **Descriptions of the program packets to be used:** Word processing, electronic mail, spreadsheets, desk-top publishing;
- **Enquiry languages to be used:** CASE tools, object-oriented languages, executive applications, image systems, expert systems;
- **Descriptions of telematic products and services,** among which are also client server concepts, videotex systems, local/long-distance

networks and voice mail systems, EDI message processors, direct dialing from the workstation.

The total end result of the IT infrastructure also indicates that the IT architecture is judged in terms of:

- *Technological feasibility*, where attention is paid to what might be feasible, in the short and in the long term;
- *Complexity*, e.g. which products and services exclude each other (possibly in the short term);
- *Controllability*, e.g. knowledge and experience of the employees;
- *Economic feasibility* e.g. cost of acquisition and development and the operating costs.

Activity 1

Name any five companies, which have adopted technology as a part of the company strategy.

- a)
- b)
- c)
- d)
- e)

9.3 USE OF IT IN STRATEGY IMPLEMENTATION

Competitive Strategy and IT

Competitive strategy is an organization's approach to achieving sustainable competitive advantage over, or reducing the competitive advantage of, its competitors. Porter (1980; 1985) suggests that the success of organizations depends on how well they cope with and manage the five key "forces", namely, the bargaining power of suppliers; the bargaining power of buyer; the threat of new entrants; the threat of substitute products; and rivalry among existing firms, which shape an industry.

Successful organizations both react to, and influence, the five forces and by doing so, influence the nature and shape of their own industry - to their own advantage, naturally in promoting growth. To fuel this growth, there are two basic commercial strategies that organizations can adopt: product differentiation (directly related to a number of the options above); and low cost/price. These two basic strategies rely for their success on a whole platform of "second order" strategies concerned with, for example, product range and distribution. They are: cost leadership; differentiation; cost focus; and focused differentiation.

Strategic cost measures which result in cost leadership are aimed at:

- Reducing the total costs of the organization by reducing or avoiding specific costs;
- Working with suppliers, distribution channels, or customers to reduce or avoid some of their costs so that the organization establishes a “preferential partnership”; or
- Increasing the cost-profiles of competitors.

If these are the activities, resulting in strategic advantage, they must be supported by appropriate technology and IT. All activities, which form part of “differentiation strategy” and which is a variant of the competitive strategy should be supported by appropriate IT. Traditional data processing relates to accounting, order processing and other administrative responsibilities. Things are now starting to change - partly as a result of changes in technology and partly because senior managers are beginning to understand the nature of their businesses and the importance of a few core processes and key tasks. Thus IT is being aimed at the frontline, customer-oriented processes and activities relating to the production, marketing, delivery, and servicing of the product.

The Value Chain and IT

All components of the value chain are interrelated. When addressing IT, it should be designed to cut across the functional boundaries and integrate the various elements of the chain. This should lead to both cheaper systems - and, much more importantly, higher quality, more strategic information through improved linkages in the chain. IT can be used to redesign and reconfigure the system by reordering, regrouping, and restructuring the activities within the value chain.

Value Systems and IT

This value chain of an individual organization, which is competing in a particular industry, is embedded in a larger stream of activities that Porter terms it “value system”. This relates to the external relationships with suppliers at one end, distributors at the other and competitors in the middle. Again, competitive advantage stems from an ability to manage these relationships more effectively than competitors. Certainly, in these days of extensive - if not ubiquitous and total - networking, interoperability of systems is vital. This extends outside of the organization so that it is certainly preferable for suppliers to share common EDI systems - and even data structures - to facilitate simpler data interchange with them. Again, this can lead to co-operative purchasing and economies of scale. It may even extend to risk management and disaster recovery agreements between organizations with similar standards and similar-sized technical installations.

For many organizations - especially small- and medium-sized enterprises (SMEs) - competitors may be a source of assistance. In many industries, co-operation with competitors is very common - through trade organizations, for example. Sometimes, small organizations have to group together in alliances to compete against a dominant large player. This cooperation may include a shared approach to, or at least experience exchange in relation to, IT.

Cooperation among organizations in relation to IT usually takes the form of (a) vertical integration, (b) outsourcing, and (c) quasi-diversification, whereby organizations cooperate across markets or across industries in order to better exploit their key resources. Adopting parallel or compatible IT means that relationships with other organizations that were previously not possible due to high coordination costs or high transaction risk may become feasible.

Occasionally, there is very wide co-operation on IT within a sector or industry. This normally relates to infrastructure components such as networking and messaging but can apply to transaction-based IT. Such co-operation may be to the benefit of all if it produces lower costs or better quality service. EDI is an example of a shared technology - offering economy of information. Few firms have investigated the issue of shared IT - though the use of packaged software is obviously a form of this. All the strategies discussed above require the organization to change. IT can be a supportive facilitator of change - extending and enhancing organization choice and improving the quality of decision making.

Activity 2

Give one example each of the following:

a) Cost leadership:

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.....

b) Differentiation:

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.....

c) Cost focus:

.....
.....

d) Focused differentiation:

.....
.....

9.4 IT FOR INNOVATION AND PERFORMANCE

Businesses are increasingly finding themselves in business environments facing rapid increases in both turbulence and complexity, leading to enhanced uncertainty and increased competition, which in its most extreme form, is termed as hyper-competition (D'Aveni, 1994). This has also led to an increased focus on innovation as a means of creating and maintaining sustainable competitive advantages (Nonaka and Takeuchi, 1995). In the wake of this development, efficiency and rapid access to knowledge and information (Barton, 1995; Grant, 1996) is becoming paramount. Consequently, spending on information technology (IT) has surged during the last few decades.

One type of benefit that managers attribute to IT is speed and responsiveness (Brynjolfsson, 1993). Speed is important in the development of successful

innovations (Kessler and Chakrabarti, 1996). There is a positive relationship between using IT to increase effectiveness and the successful implementation of innovations. Grant (1996a; 1996b) focuses on knowledge integration within companies as important for creating sustainable competitive advantages. Huber (1990) further argued that the use of IT leads to more available and more quickly retrieved information. The importance of knowledge integration and availability of information may be linked to the positive relationship between using IT to improve internal communication and successful innovations, facilitating a higher degree of coordination and integration of activities. Also, by using IT, it is likely that the speed of the processes increases, leading to lower costs of development and providing an earlier introduction to the market (Kessler and Chakrabarti, 1996), which in turn will have a positive effect on the successful implementation of innovations. Use of IT can also lead to new ways of managing the company and enhancing productivity. This can be explained by the fact that IT has an inherent potential in terms of improving coordination of cost reducing activities in the company and thus raising cost efficiency.

Customer Satisfaction

Berkley and Gupta (1994) found a positive relation between the use of IT to improve service quality and customer satisfaction. This was also found by and pointed out as being of significant importance for many firms by Quinn (1996). The positive link between the use of IT to ease the work and customer satisfaction may be attributed to the inherent time saving, which could be transformed into more attention directed towards the customers, thus increasing customer focus and in turn customer satisfaction. The positive relation between the use of IT and lower costs, and customer satisfaction may be caused by customers benefiting from cost reductions in the way of price reductions, which will lead to increased customer satisfaction.

9.5 E-BUSINESS

The use of the term e-business implies that it is distinct from business *per se*. There have been arguments proposed that would suggest that the underpinnings of e-business are of such significance that it can be regarded as discontinuously different. If this is so, then it could be argued that a new business paradigm has emerged. The recent reversal of fortunes of the dot.com businesses has graphically emphasized the fact that they, and e-business, operate within the same business environment and context as conventional bricks and mortar businesses (Porter, 2001). However, fundamental to this is the technology that has enabled the e-business phenomenon to take place.

Web-based Business Models

The rapid expansion of the worldwide web and e-Commerce has created a number of new business paradigms. The array of business relationships which have emerged include:

- Business-to-Business (B2B)
- Business-to-Consumer (B2C)
- e- Market Places

The business-to-business space includes the various upstream and downstream transactions that enhance channel coordination and customer relationships. In B2B commerce, business partners and customers are connected via the Internet to participate in commercial trading and participate in communications and interaction. JC Penny, for example, shares packing, shipping, inventory and product movement with suppliers. In B2C, business are connected to their customers through the net and the various activities in the B2C space include, product ordering, sharing product information, creating display space, providing customer information, co-developing products and customer service. An example of B2C commerce is the product-tracking information offered by Federal express to the customers through its B2C activities. E-Market Place involves linking the company, its partners and its customers via the Web to provide opportunities for developing communication and interactions, including customer surveys and information exchange on such things as product warranty and service capabilities. Figure 9.1 explains the web models and this interrelationship.

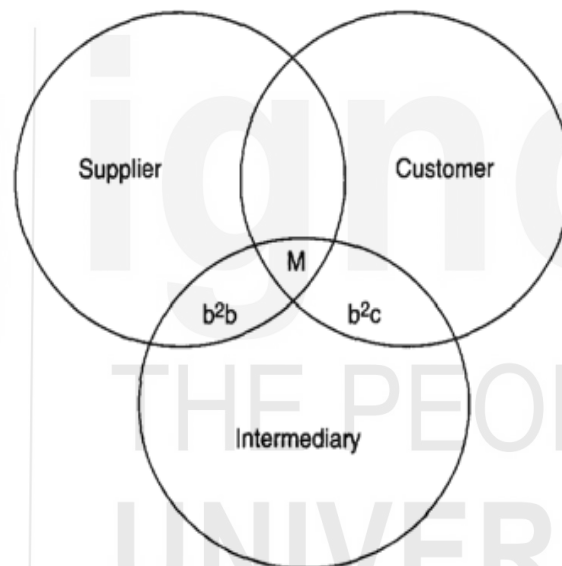


Figure 9.1: Web-based Business Models

In the context of e-business the business to consumer sector (B2C) can be considered as secondary to the potential within the business-to-business (B2B) sector. For example, one of the largest supermarket businesses in Europe, Tesco, also has the largest B2C e-business in Europe and one of the few profitable ones. However, despite this the revenue from this business is only approximately £300 million, from a total turnover of £23 billion.

The opportunity to link B2B network members and co-ordinate their activities by the quicker and more effective transmission of information relating to stock, order, administration, etc. has considerable potential for cost saving and service improvement. The often-quoted example of Wal-Mart and Procter & Gamble suggests that early and more immediate benefit can be gained by working through and interconnecting the supply chain (Christopher, 1998). Again, Tesco offers a further example with their leadership in this context using TIES (Tesco information exchange system) to link with their suppliers. The company also leads a forum of retailers with the aim of standardizing systems used by retailers and their suppliers.

Impact of e-Business on Organizations

The diffusion of Internet-based information systems throughout the workplace is changing the manner in which the firms work and the way they interact with their customers and suppliers. According to McKenna (1997), technology is transforming business environment in profound ways. Almost all technology today is focused on compressing to zero the time it takes to acquire and use information, to learn, to make decisions, to initiate action, to deploy resources, to innovate. When action and response are simultaneous, the firms are said to operate in real time.

This shift in time-demands and resulting competitive pressures are changing organizational structures: Increasing competition, information that could be outdated in the next hour and customers demanding immediate responses to product and service requests (customized to their needs) mean a major adjustment in company structure. To meet the challenges effectively, many companies have adopted reorganization strategies that include downsizing, restructuring, re-engineering, outsourcing, and merging or spinning off companies.

The transition of operations in many businesses to real time operation has begun to churn up huge creative chaos inside corporations (McKenna, 1997). The Twenty-first century Corporation must adapt itself to management via the Web. It must be predicted on constant change, not stability, organized networks, not rigid hierarchies, built on shifting partnerships and alliances, not self-sufficiency. Internet-based technologies are creating information overloads where “a major task of practitioners is to help mobilize those frozen by the overload of information”. Web-based information systems are also causing profound organizational changes: It seems logical to assert that Internet-based information systems have created profound changes throughout organizations. Researchers have begun to document the impact of these changes. One common thread appears to be that applications should be Web-based in their design, and that they support distributed collaboration and decision-making.

Organizational Change

In most organizations, few people possess all the information required to make optimal decisions. By taking advantage of Web-based information systems, associates can access information from many sources at any time and from any place. This has created more team-based decisions and new organizational structures. Thus, the inference is that Web-based information systems can lead to organizational changes by facilitating organizations that are more organic in structure and can adapt more easily to share information in a timely and coordinated fashion. Web-based technologies have similarly created performance improvements in organizations by changing roles and patterns of communication. Electronic networks open up new possibilities for reducing barriers to communication and sharing organizational knowledge and electronic collaboration tools can tap into expert knowledge and resources throughout an organization where productivity, flexibility, and collaboration will reach new, unprecedented levels. Success in increasingly competitive marketplaces will depend on effective communications and knowledge sharing among members using collaborative electronic networks.

Time

In terms of the impact of Web-based information systems on time, Byrne asserts, “Employees will increasingly feel the pressure to get breakthrough ideas to market first”. He further contends that rapid flow of information will permeate the organization. Orders will be fulfilled electronically without a single phone call or piece of paper. The ‘virtual financial close’ will put real-time sales and profit figures at every manager’s fingertips via the click of a wireless phone or a spoken command to a computer”. He further contends, “It is about speed. All this work will be done in an instant”. “The Internet is a tool, and the biggest impact of that tool is speed”, says Andrew S. Grove, Chairman of Intel.

The speed of actions, the speed of deliberations, and the speed of information have increased, and it will continue to increase. That means the old, process-oriented corporation must radically revamp. With everything from product cycles to employee turnover on fast-forward, there is simply not enough time for deliberation or bureaucracy. While this sounds like a positive impact, the shrinking time demands may cause increased problems in other areas, for example, employee stress.

Web-based information systems will have a profound impact on the organization and its structure. Organizational chart of large-scale enterprise had long been defined as a pyramid of ever-shrinking layers leading to an omnipotent CEO at its apex. The twenty-first century corporation, in contrast, is far more likely to look like a web: a flat, intricately woven form that links partners, employees, external contractors, suppliers, and customers in various collaborations. The players will grow more and more interdependent. Fewer companies will try to master all the disciplines necessary to produce and market their goods but will instead outsource skills - from research and development to manufacturing - to outsiders who can perform those functions with greater efficiency.

Therefore, as Web-based information technologies diffuse throughout organizations, there may be profound impacts on organizational changes, including the general flattening of organizational structures and the need to develop middle management teams that operate effectively within this new environment. Web-based electronic networks have been shown to have both positive and negative consequences. Anticipated, desirable consequences have included timely savings, improved productivity, and improved decision making via increased access to timely information. Innovation and creativity were also shown to improve when workers could share ideas and knowledge. On the other hand, researchers have demonstrated that Web-based information systems can also have a negative impact on workers. People feel pressurised by the real-time demands created by the non-stop presence of the Internet. They also sense loss of communication and relationships created by virtual communities and meetings, relationships based on physical and face-to-face meetings and conversations.

Identifying e-business Application Areas

When examining the company for the possible application of e-business, one can focus either on internal processes and systems or on the externally oriented processes. If the main focus is to reduce costs or prepare systems for

future e-business applications, the internal perspective might fit best. If the aim is to improve the customer's perceived value, one can best investigate the company's buying and selling processes.

Internal e-business Value Chain

Taking the value chain (Porter and Millar, 1985) and placing e-business technologies into the framework gives an insight into the reach of these technologies into the value activities. The exact meaning of all prevalent "e-" applications is less relevant as new applications arise every day and definitions vary widely. Linkages already exist between activities; some of these linkages have been integrated by using e-business technologies, ultimately providing a fully integrated e-business process. It is important to realize that these new applications have to be integrated with supporting and, if applicable, primary processes to prevent creating islands of automation.

The physical processes might have to be rearranged to better align the original value chain to the new e-business oriented value chain. Integration of the physical processes and e-business applications is essential to achieve maximum results. It is said, "A business is profitable if the value it creates exceeds the cost of performing the value activities" (Porter and Millar, 1985). Analyzing the e-business value chain can help in lowering the costs and increasing the value of activities. It has to be kept in mind that the supporting processes should be prepared for future e-business developments before embarking into large-scale "e-" systems.

Taking the Web marketplace as an example, one can see that, if a marketplace requires sound estimates for the delivery time of a product, e-fulfillment systems have to be in place and the factory floor automation has to be capable of providing this information. Supporting processes are not only the technical infrastructure, but also the databases holding all information and people capable of working with the systems.

Formulating an e-business Plan

Having identified the portfolio of specific e-business applications that need to be developed from a strategic perspective, these applications have to be brought into line with the existing IT architectures. Commonly identified IT architectures encompass the following:

- Information Architecture;
- Systems Architecture;
- IT Infrastructure; and
- Organizational Architecture.

As a first step, the impact of the identified e-business applications on the information architecture has to be assessed. The e-business applications can be integrated into the information architecture, taking the customary view of information architecture as the description of information systems areas in terms of the business processes they support and the data they use. Three possible situations can occur at this architectural level:

- 1) An e-business application fits well within one information systems area. This means that the original information architecture is still valid.

- 2) An e-business application covers two or more information systems areas. A decision needs to be made whether to merge the affected areas or to rearrange them in such a way that the e-business application falls into one new information systems area, together with possibly existing or projected other applications. The relationships between information systems areas have to be redefined in order to arrive at consistent information architecture for the new situation.
- 3) An e-business application does not fit in any information systems area. A new information systems area has to be defined in terms of the business processes supported and data items created and used by the e-business application. Careful analysis of possible relationships with other information systems areas within the information architecture is needed.

A similar process has to be followed through for the systems architecture. Specific attention has to be given to the stewardship of the data items (who controls the creation of data, which mechanisms have to be in place to control any occurring redundancy?) and the integration of applications (how can we provide a consistent interface to the users?). The consequences for the IT architecture might be more severe, as e-business applications often call for substantially higher degrees of scalability and security. Accordingly, the required capacities and skills for the supporting organizational architecture may be very different from existing ones, which often gives rise to a renewed outsourcing discussion.

Through assessing the impact of e-business applications on existing architectures, several consequences are identified. Plans have to be made to properly incorporate these consequences within the IT architectures, describing what changes are needed to existing information systems, infrastructure, new developments or organizational layout. These consequences give rise to projects within the overall IT project portfolio of the organization. As a result, the project portfolio is populated with both e-business application projects and projects that need to be carried out in order to properly integrate the e-business applications with the business structures and IT architectures of the organization. Standard project portfolio management techniques can be employed to render a specific e-business plan for the organization.

9.6 IT IN SERVICE SECTOR

Past investments in service sector information technology have been aimed primarily at productivity or efficiency gains, and this is the measure that most firms use to gauge the benefits. Service firms have followed the lead of manufacturers in making great strides in getting work done with fewer employees mainly because of advances in technology. Competitive pressure is making service companies eliminate costs (mostly people) and new easy-to-use software is allowing the full application of computer power. On the other hand, little attention has been given to using information technology to improve customer service and long-run business effectiveness. Perhaps this is because the benefits of improved service are often qualitative rather than quantitative. Standard accounting systems can measure labour costs,

but not the costs of poor customer service. Consequently, managers must justify new investments in information technology on the basis of efficiency gains and labour savings (Berkley and Gupta, 1994).

Input Information

The input function in services includes forecasting customer demands so that necessary service capacities can be planned. Once customers arrive, expected services must be specified by questioning customers or by relying on service histories or observations of market trends.

Most service firms have rush or peak periods and are not able to provide quality service unless they plan and prepare for these times. Unlike manufacturers, service firms cannot maintain inventory of their products as a hedge against fluctuations in demand. At any given time a service may have excess demand or excess capacity and service quality can suffer in both cases. When demand exceeds capacity some potential customers may be disappointed because they are turned away. There is also a risk that the accepted customers may receive inferior service. For many customers, even if it is good service, it is no good when it is late or slow. Armed with the proper information, service firms may be able to adjust capacity to match fluctuating demand levels. When peaks in demand are predictable, forecasting and capacity management systems can be used to construct detailed staff schedules that match capacities to demand.

In services, information must be secured from the buyer to specify the expected service. This is important because the more complete the information, the easier it will be to perform the other process functions. Customers also need to be made aware of the various services available and the likely costs of each alternative. Such information ensures that the needs and expectations of the customer are fulfilled and the organization's time and resources are not wasted in dealing with customers whose needs and expectations it cannot, or should not, fulfill.

Service requires a long memory. With a computerized customer database, a firm can attach a detailed personal service history to the names of its customers. A record of each new service transaction can then be added to existing customer files. These updates help sketch an increasingly detailed profile of each customer's preference and expectation and create opportunities for more personalized and enhanced service. Customer service histories that are easily accessible allow frontline service providers to know on the spot which customers are first-time clients and which are loyal repeat customers. Such information allows service staff to acknowledge and personally reward the valued repeat customer and to solicit feedback and other important information from new customers.

Customer files enhance service consistency and server competence. Customer service records also ensure that service is personalized and consistent for repeat customers, even after their regular service-delivery person moves on to another job. The Nordstrom department store chain depends on its sales associates to provide individualized service to its loyal customers. Currently, individual customer preferences are resident only in the memory of a sales associate and the firm is working to convert this personal memory to corporate memory. An information system that allows

customer files to be called up at many different locations would allow the firm to direct customers to different company stores providing individual sales or services of special customer interest. This in turn will help build a customer-company relationship that is stronger and more valuable than a simple customer-store or customer-employee relationship. Customer service histories can speed service. In the medical field, computerized patient records speed service, cut costs and save lives. For example, computerized systems can warn physicians of potential problems such as allergic reactions or duplicated tests.

Customer service expectations are a moving target. To deliver superior service, a company must monitor customer expectations and customer response to the services it offers. While market research can be used to determine customer expectations, often the required information can be obtained at a significantly lower cost by listening to customers and employees. Most of the good service providers have a communication process to ensure that customer suggestions and requests are communicated up and down the organization to the people who need this information.

Although customers are the best source of information, they will rarely volunteer the necessary information. In industries characterized by large numbers of relatively small transactions, such as financial services and retailing, computerized point-of-sale and bar-code scanning devices can now record every customer service encounter. For example, in supermarkets, scanners speed checkout and provide customers with detailed receipts. Moreover, scanner systems provide management with continuous inventory updates and a detailed analysis of performance by product, by department and by store. The intended result is fewer inventory stock outs of popular items and improved customer satisfaction.

Knowledge

Service providers must possess the required skills and knowledge to perform service. Greater knowledge allows frontline service workers the better to help their customers and makes them capable of important judgements on matters that previously would have been handled by managers. Because employees can experience intense frustration when facing a customer and not having the answers, knowledge also supports employee job satisfaction, motivation and confidence in dealing with customers.

Knowledge databases allow relatively inexperienced people to perform very sophisticated tasks quickly. Whereas service providers, unaided by databases, are limited to their own knowledge, those with access to fast-response decision-support systems effectively possess the knowledge of many. This is particularly important when service firms rely on entry-level, part-time or relatively inexperienced workers. Information systems can also be used to reduce the knowledge required to deliver customized services and to improve service consistency. If the most relevant customizing variables can be specified and programmed in advance, the firm becomes less dependent on frontline personnel to perform the customizing tasks. For example, employment agencies try to identify job openings offering the desired salary, location, type of work and level of responsibility. Computer programmes can be written to search the job-opening files and automatically

generate a list of feasible matches. Not only is the market knowledge institutionalized, but also much of the necessary expertise.

Quality in services depends heavily on the ability of employees to share their knowledge. Service expertise can be captured in either expert systems or group conferencing systems that provide electronic bulletin boards for sharing problems and ideas. Many professional service firms now find the core of their distinctive competence to lie in the accumulated knowledge in their databases and the capacity of their members to access and build solutions on these databases. A number of software companies maintain textual databases of reported software problems and solutions. As solutions to particular problems are found, they are added to the database and become widely available to the technical support staff that takes calls from users. By recording problems and solutions centrally, these databases give leverage to the learning of each technical support person.

Job Status

The longer it takes for service delivery to be completed, the more likely it is that customers will require information on work-in-progress (such as estimated completion times and projected costs). Many firms have developed customer information systems that allow customers direct access to production and shipping files. These systems reduce customer uncertainty and allow customers to measure firm performance. Frequent airline passengers expect occasional delays. What upsets these passengers is the lack of explanation and apology for delays. To be more responsive, one of the airlines passes information from its flight monitoring system to co-coordinators located in each airport who make sure passengers know the reasons for delay.

Quality Control

Quality control consists of collecting data, monitoring (comparing the existing state with the service standard) and corrective action. The objective is to make corrections to the process before problems are created and customers complain. Many service problems can be identified before customers experience them. Consider patients who arrive at their doctor's office on time only to be told the doctor is running an hour late or airline passengers who, on arrival at the airport, are informed that their flight was cancelled hours earlier. In situations like these, management could anticipate customer frustration and take steps to alleviate it, including calling customers to warn them of the problem.

Quality control begins with data collection to determine the current state of the process. This information is then compared to the service standard to determine if corrective action is required. When service standards are subjective (e.g. courteous service) or when the data are qualitative (e.g. employee behaviour, customer treatment, customer reaction), qualitative data are ordinarily collected by direct management observation. On the other hand, objective performance data, such as customer waiting and service times or system response times, can be collected and processed by information systems.

Complaints Management

Customer complaints provide valuable information regarding service quality problems. A problem resolution situation should be viewed as an opportunity to learn how to improve service. The greatest risk is that customers will not bother to complain, but will simply generate negative word-of-mouth advertising and take their business elsewhere. Service firms should welcome complaints and make it easy for customers to complain.

The closer to the point of service delivery a complaint is made, the better is the service recovery. Experience in many companies indicates that it takes longer to handle an escalated complaint at the head office than it does at the point of service. Once a complaint is lodged, fast response is the key. Customers should not have to wait weeks to get an answer or to get a problem resolved.

Successful service firms track complaints by type (e.g. poor employee attitude, slow service), by frequency and by department. This is done because many service problems are not so obvious and, without adequate tracking systems, often go undetected. Some service companies also use complainant satisfaction tracking systems to measure the success of their complaint handling systems. These systems generally send customers who have complained a postage-paid reply card for evaluating the way their complaints were handled. Customer replies can then be tabulated by individual customer service representative, by location or by teams of complaint handling personnel.

Customer feedback is not always bad. Service firms also receive compliments. Customer compliments provide an opportunity to increase employee motivation and improve service quality. Unfortunately, many companies do not have an organized system for routing compliments back to employees. This is particularly true for geographically widespread organizations where a compliment might be received in Singapore about service delivered in Paris. Verbal compliments should be recorded (the format is not important) and, with written comments, passed on to all employees who contributed to the service complimented and to their immediate supervisors. Typically, the effort and money spent on using compliments to motivate and encourage superior performance are returned many times over.

Service Recovery

The best service is preventive rather than reactive. But, despite one's best efforts, mistakes are a crucial part of every service. The fact is, in services - often those delivered in the customer's presence - errors are inevitable. But dissatisfied customers are not. A good service recovery can turn angry, frustrated customers into loyal customers. Good recoveries can, in fact, create more goodwill than if things had gone smoothly in the first place.

Service failures are best resolved when and where they happen, before they become costly to resolve and before they create lost revenue. To resolve problems when they occur, frontline personnel must be trained and encouraged to use their judgment. Employees need enough data to solve problems and make decisions while the customer is still present. In many

cases (such as billing problems), recovery efforts require customer account histories and data from several company departments. If problems are to be solved on the customer's first call, this information must be readily available to customer service personnel.

Nothing could be worse than saving a customer only to have the same mistakes repeated because other employees who needed to know were not informed. To prevent problems recurring, and to prevent weak recovery efforts that fail the customer twice, some firms use recovery-tracking systems that capture information pertaining to each instance of recovery service. This information is available so that all employees who deal with a particular customer will know what occurred, what recovery methods were used and what commitments were made. For instance, if a restaurant seats a patron with a reservation very late and promises a free dessert, the waiting-on staff should not later add this dessert to the customer's bill. To ensure accurate data, customer service representatives should be able to input information directly into the recovery tracking system. Direct access also facilitates retrieval of information helpful to recovery efforts.

Customer Defections

Measuring service quality objectively through conformance to standards and subjectively through customer surveys is not enough. These techniques miss former customers who have left over the company's handling of an irregular situation. Identifying defecting or lost customers and measuring defection rates can provide a way to measure and improve service quality. The idea is to identify those customers who stopped doing business with the firm, then find out why. Defections can then direct managers' attention to the specific things that are driving customers away.

To measure defections one must have a defections scanning system to identify customers who have ended their relationship with the firm. If service or billing histories of customers are available, scanning the dates of last account activity easily identifies defections. Alternatively, many service firms, such as airlines, hotels, restaurants, rental car agencies, retail stores and even grocery stores, now have membership programmes and customer databases.

Often, customers are given a membership card that entitles them to discounts, and all subsequent purchases are logged against the card number. These databases then provide service managers with an easy way to identify inactive customers and, often, clues as to why customers are no longer buying.

9.7 SUMMARY

The rapid growth of technological innovations and the synthesis of information technology and computer networks are radically changing the way companies compete. Many business enterprises are making strategic commitments to technology for the purpose of gaining and sustaining competitive advantage in their industry. The creation of a competitive advantage through the use of information technology (IT) requires business executives to control this vital corporate resource and manage its use.

Organizations seek to gain competitive advantage in their selected markets via a number of competitive approaches - based primarily on product, service differentiation and pricing policies. To understand the environment and customer behaviour they need robust, reliable information. In order to deliver to their chosen strategy, they must configure the organization (including extended configuration within the industry sector through alliances and collaborative ventures) and the various functional processes to deliver reliably and efficiently. Information Systems (IS) is used to configure the organization appropriately and to ensure communication between the various components. IS is then used to ensure effective communication within the extended value chain involving suppliers and the distribution network. Organizations hoping to make strategic use of IS must understand the nature of the inter-relationships of the above elements, and understand the nature of the information flows between them. IS activity and expenditure should be prioritized where it is clear that strategic information will result or where the activity is not effective without an underpinning IS.

Organizations, which are at the beginning of the use of IT often, focus on efficiency. They can do without a formal information planning method. The periodic drawing up of priorities and action plans per system is often sufficient. In this organization, IT is often placed in the existing procedure, without great changes in assignments or in the organization structure. In a later stage, automation is applied to solve bottlenecks, in the information supply and to improve effectiveness. For this purpose, it is necessary to chart the company processes and the information needs. The (often functionally designed) process structure of the organization usually remains intact. Most of the well-known information planning methods are extremely suitable for this type of information planning.

Increasingly more organizations, because of their complexity and the complexity and competitiveness of the market, are compelled to function in a more client-oriented way. In this case, information supply without bottlenecks is a necessary, but not a sufficient condition. It is even more important to choose the right strategy, process structure and responsibility, and in association with these, the right set of applications and the right infrastructure. These organizations discover that the functional organization structure involves obstacles and that information planning should not merely involve the establishing of priorities for individual system development projects.

In high customer-contact services, a firm's ability to deliver quality service depends on its capacity to collect, process and distribute information. The input function in services includes assessing customer expectations, specifying the expected service and setting corresponding service standards. Good service providers have communication processes to facilitate the collection of customer data, suggestions, requests and transactions into customer databases. These databases can then be used to construct detailed customer profiles, eliminate service-specification errors, speed service and improve service consistency. Output information is used to determine whether customer expectations are met. While customers are the best judges of quality, many service firms lack adequate systems for collecting and acting on customer data.

Customer complaints provide valuable information on service quality problems. If customers complain, employees need enough information to solve problems and make decisions while the customer is still present. Complaints should be tracked by type, frequency and department to identify recurring problems that otherwise might go undetected. To prevent weak recovery efforts that fail the customer twice, some firms use recovery-tracking systems to capture and distribute information pertaining to each instance of recovery service.

9.8 KEY WORDS

IT Services: Services which execute specific assignments for applications.

IT Control Instruments: These instruments consist of procedures, methods, techniques and tools for system development and the quality and experience of the automation personnel.

Competitive Strategy: It is an organisation's approach to achieving sustainable competitive advantage over, or reducing the competitive advantage of, its competitors.

9.9 SELF-ASSESSMENT QUESTIONS

- 1) Explain the role of information Technology (IT) in strategy implementation. How can IT assist in enhancing the competitiveness of a firm?
- 2) What are the various components of IT architecture? What factors influence the choice of a particular IT infrastructure?
- 3) What is e-business? Briefly explain the various web-based businesses? Explain the steps involved in implementing an e-business plan?
- 4) How does IT improve innovative capacity and performance of a firm? Illustrate this with an example by scanning various sources of information such as web, journals, business dailies, etc.
- 5) IT is being extensively used by various service organizations to improve service delivery. Choose any service organization and explain how IT has enhanced the quality of service of this firm.

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UNIT 10 TECHNOLOGY AND R&D

Objectives

After reading this unit, you should be able to:

- highlight the importance of technology and R&D to the competitiveness of a firm;
- describe the features of technology and acquisition of technology;
- understand technology search strategy and challenges of technology audit;
- assess the contribution of R&D activities to the competitive strategies of the firm;
- discuss the steps involved in developing R&D Strategies; and
- state the reasons for limited progress made by firms in R&D Strategy.

Structure

- 10.1 Introduction
- 10.2 Technology and R&D in Organisations
- 10.3 Features of Technology Package
- 10.4 Competitive Strategy and Competitiveness
- 10.5 Competitive Advantage and R&D
- 10.6 Value Chain and Value Chain Analysis
- 10.7 Development of R&D Strategy
- 10.8 Steps Involved in Developing R&D Strategy
- 10.9 Progress of R&D Organizations in Strategy Development
- 10.10 Summary
- 10.11 Key Words
- 10.12 Self-Assessment Questions
- 10.13 References/Further Readings

10.1 INTRODUCTION

Technology has a strategic connotation. The technological strategy and its proper management enhance the competitive advantage. Technology helps in building of knowledge and skill. Technology can be studied in the following forms. There are three types of technology. **Product technology** refers to the various features of the product. The knowledge needed for processing or manufacturing a product is the **process technology**. The skills required to run a business is **management of technology**. Prior to development or acquisition of technology it becomes necessary to identify the technological gaps and technological competence to fill the gap.

An essential component of competitive strategy is recognizing the role that Research and Development (R&D) plays in the competitive success of a firm, and acting to ensure that technology decisions and policies contribute to the

firm's competitive advantage. This unit provides a framework, which can be used to analyze and understand the importance of technology development, acquisition of technology, technology audit, the linkages between R&D and competitive strategy and/or competitive advantage of a firm.

10.2 TECHNOLOGY AND R&D IN ORGANISATIONS

Technology development includes both R&D and development aspect. Technology generation and development is often synonymous with the term research and development (R&D). However, technology generation involves R&D efforts while technology development involves further stages of translating R&D efforts into marketable products, processes and services. Recognition of a need for innovation is one of the motivations for R&D. "Research" on existing knowledge for satisfying identified need helps in idea generation – this is the "need push". The other primary motivation for R&D is to find potential applications for advances in knowledge. "Research" on existing activity for introducing new knowledge also helps in idea generation – this is the "technology-push". "Development" includes engineering (creation, design and production) and marketing (first use and diffusion) of the generated ideas. Through the entire process it is ideas and knowledge which are being pursued, and the process is not complete until the new idea is converted into a marketable product or service (a hardware or software intensive technology).

Technology is generated in R&D organizations. Goals, surroundings, criteria and resource allocation are some of the inputs to R&D, the output of which is technology. The input resources into R&D organizations are the traditional inputs such as money, materials, facilities, energy, labour and management, and the intelligence based inputs such as science, knowledge, skills, information and existing technologies. The effectiveness of any R&D is determined in terms of the 'usefulness' of the technologies it produces with respect to the overall objectives of the corporation. The R&D or technology generation involves many other aspects such as, monitoring and evaluation of R&D projects, funding of R&D, training and development of resource personnel, interactions at all levels, management policies and support, the availability of support structures and incentives at government level, timely collection and interpretation of technical and other information, etc. Thus technology development should be cost effective and fulfil consumer preferences.

10.3 FEATURES OF TECHNOLOGY PACKAGE

The technology package consists of three principal elements namely, product design, production technique and management systems.

Product design may range from simple items to highly complex (e.g., automotive) parts. **Production techniques** and plant layout include blueprints and flowcharts, formulas and recipes, process sheets, fabrication instructions, tools and fixture designs, operational procedures and material specifications. **Management Systems** consist of various plans, layouts and technical control systems (along with related marketing and financial

controls). Included are plant design and layout, quality control and testing, material procurement, inventory control, equipment maintenance and repair and machine loading techniques.

The three principal categories of technical information or know-how inherent in technological systems are general knowledge, system-specific and firm-specific knowledge. These various categories of knowledge may be in the form of written materials or may be embodied in technical assistance, on-the-job training or built into fabricating or processing equipment.

The concept of technology absorption differs from country to country, and even from firm to firm. In India, absorption is generally considered as the capability to reproduce or manufacture products according to the “know-how” supplied by the licensor of technology. The need for technology absorption at the firm level in India, and the measures being taken to encourage absorption efforts are to be highlighted.

Adoption

Adoption of a technology is a process under which the various features of the technology which is the subject of transfer are suitably modified, changed or altered keeping in view the needs of the buyer.

Adaptation

Adaptation of technology is a phase that takes place after a technology has been adopted and put to use in production activities/facilities.

Absorption

Technology is said to be absorbed if it is fully understood, so that it is in a position to be further optimized and upgraded. Technology absorption involves ‘know-why’ exercises, basic investigations into the product and/or process and/or systems.

Acquisition of Technology

Technology can be acquired through purchase or developed through research and development.

Optimization

After understanding the relevant features of technology, further exercise in ‘removing rough edges’ through R&D and value engineering to effect savings in the use of material and energy consumption, etc. both in product and processes constitute ‘optimization’ of technology.

Improvement and Upgradation

Capability in technology absorption and optimization can lead to further exercises in improving the existing products and processes by R&D efforts of industry and other associated research organizations. This will enable industry to meet the changes in technology of the product or processes. Technology upgradation exercises lead to industry’s efforts in extending its know-why capability to a higher range of products or in upscaling the existing process/production technology or manufacturing equipment.

Technology Evaluation

Technology evaluation as a formal tool has not been applied or used in most cases of technology acquisition in India. For exports not only product

specifications and quality but also manufacturing practices and facilities would need certification and accreditation. This would require that the entrepreneurs carefully select the technology to be adopted.

Protection of Technology

The innovated technology needs to be protected in order to protect the competitive edge. Patent and trade marks are two such methods. Temporary monopoly rights are provided through patents while trademark is the unique work of the product which gives the product a distinct identity and provides protection from similar products of a rival firm.

The above measures need to be adopted to formulate a technological strategy and avoid rivalry thus safeguarding the uniqueness of the product.

Technology Search Strategy

A company, big or small, may at one time or other, require transferring technology or import of technology from outside. The process of transferring technology either 'in' or 'out' is subject to both managerial and other resource limitations. Technology search strategy has to be undertaken by the unit to identify suitable technology within the enterprise or import of technology from outside to maintain growth and profitability of the company.

The market conditions in any country are dynamic and can operate in a very ad-hoc manner. It should be a major concern for companies to undertake search strategy to identify suitable projects or components for sustained growth. An effort to find a suitable new product and knowledge of the potential licensor of that product may lead to an early decision and successful implementation of the project.

It is useful to define why new products are required, the type of product that is required, its stage of development and whether this product will fit the existing skills and resources within the firm. The success of seeking a new product will also depend on, among other factors, their technology search strategy and whether the relevant factors are defined and employed at the outset.

Technology audit caters to three broad areas. Firstly, the risk related to use and acceptance of technology and the duration of relevance of a particular technology. Secondly, it is the firm's competence to develop or acquire the relevant technology and the ability to commercialize the technology. Thirdly, every technology requires research and development. The challenge with technology audit is to identify the amount of investment that will be needed in the development of technology.

10.4 COMPETITIVE STRATEGY AND COMPETITIVENESS

The conventional approach to strategy has emphasized setting goals and developing the means to achieve them by matching the resources of the firm (strengths and weaknesses) with opportunities and threats in the external environment, which includes, especially, customers and competitors, and deciding which industries, businesses, or product-market segments to compete in. At the highest corporate level, there is the multi industry-

business firm, conglomerate or the diversified firm, such as Reliance Group, Tata Group, etc. At this level, corporate strategy addresses issues like as choosing a balance among the industries or businesses chosen, and -in the case of a firm- achieving synergy among the industries or businesses chosen.

Moving down a notch, there is the single industry-business firm such as -- cement, or the division of the multi industry-business firm such as Aditya Birla Group. At this level, the issue of which business or industry to be in is important and the only issue is how to enter or exit from a business or industry. This is the level at which competitive strategy operates. Developing a competitive strategy essentially involves building a broad framework for a firm on how it is going to compete, what its goals should be, and what policies will be needed to carry out those goals. Moving down to the next level, there are the functions of the firm-- the engineering, manufacturing and production, marketing, sales, service, personnel, human resources, purchasing, accounting, finance, planning, etc. The term 'functional strategy' is widely used at this level. The important point to note is that functional strategies must all support, reinforce, and contribute to the competitive strategy of the firm in order for the firm to effectively compete.

In a free-market economy, the generic goal of any firm is to enhance its competitiveness, which can be defined as the ability of a firm to get customers to choose its product or service over competing alternatives on a sustainable basis. Competitiveness is measured by market share trends over time, and can be described in terms such as increasing, decreasing, or stable. There means an important stipulation contained in the definition described above, however, which is that competitiveness must be built *on a sustainable basis*. It is possible, in the short run, for a firm to get customers to buy its products or services over competing alternatives on an unsustainable basis by, for example, mortgaging its assets and using the proceeds to subsidize and lower prices, thus attracting customers until the earnings run out and the firm collapses.

10.5 COMPETITIVE ADVANTAGE AND R&D

A firm can enjoy competitive advantage in several ways. For instance, a firm may gain competitive advantage because:

- The *price of its product* is lower.
- The *quality of its product* is higher.
- *Availability of its product* is sooner, or more dependably just in time.
- *Customer service* is better.
- *Attractiveness of its product* is greater.
- *Awareness of its product* is greater.
- Other social, psychological, and ideological factors.

In practice, customers shop around amongst competitive alternatives by setting parameters for some of these advantages and then making the final choice on the basis of the key or critical advantage. Customers today expect high reliability and low prices and these are mutually reinforcing attributes that a supplier is expected to achieve just to be in the competition.

But these are not often enough. The winning competitor must have both the lowest price and highest reliability or achieve one of the other competitive advantages that customers value. A firm, which has a strong R&D programme, can influence all these factors positively and contribute to the competitive advantage of a firm. Competitive advantage views R&D activities as a way of improving a process, thus reducing costs, or providing customers with the best-of-class benefits.

Porter defined and discussed three types of generic strategies: cost leadership, differentiation, and focus. Low cost leadership means essentially what it says ; achieving the lowest-cost position possible in each operation of the firm, not just manufacturing, through such means as vigorous cost reduction programmes, strict cost and overhead controls, economies of scale, and learning curve efficiencies. Differentiation refers to unique feature of a product or service as perceived by a customer, which directs the customer to prefer it over competing alternatives. A differentiating uniqueness is typically achieved through such means as design features, establishing a brand-name identity, or offering superior customer service. The linkage between the earlier list of competitive advantages and Porter's strategies should be obvious; low-cost leadership corresponds to a potential competitive advantage; differentiation corresponds to uniqueness in terms of quality, sooner availability, better customer service, etc.

10.6 VALUE CHAIN AND VALUE CHAIN ANALYSIS

Till now the unit has dealt with competitive strategy by emphasizing customers and their ultimate role in determining competitiveness by collectively choosing to buy the products or services of one firm over competing alternatives. In this section the focus shifts to firms and what firms do to achieve competitive advantage in implementing their competitive strategies. To address this set of issues, the value chain analysis, a tool developed by Michael E. Porter is utilized. According to Porter, the value chain is "a systematic way of examining all the activities a firm performs and how they interact for analyzing the sources of competitive advantage."

The value chain can be used to organize all the activities of a firm into categories of primary and support activities. Value Chain Activities are processes or things that are done in a firm. In order to be identified and categorized, they must be distinct; they must have a beginning and an end, which distinguishes them from other activities or operations. Primary activities constitute the processes by which firms receive inputs (in-bound logistics), convert those inputs into outputs (operations), get those outputs to customers (out-bound logistics), persuade customers to buy the outputs (marketing and sales), and support customers in using the outputs (service). Support activities are processes which provide support to the primary activities and to each other in terms of purchasing inputs (procurement); improving existing products or developing new products (research and development); dealing with personnel (human resource management); and general management, accounting, finance, and other activities which support the entire organization rather than individual activities (firm's infrastructure).

Using the value chain and focusing on categories of activities enables one to see the firm as a collection of activities rather than as an organization chart and administrative units such as the purchasing department, the R&D, the manufacturing division, and the personnel department. This is important because the value chain activities be analysed much more effectively by using the value-chain analysis, firms can then create competitive advantage in the following three ways:

- 1) By placing greater or lesser emphasis (allocation of resources, management time and attention) on specific activities than competitors do.
- 2) By performing specific activities better (better management, more highly trained people, better-maintained equipment) or differently (using an alternative-- presumably new or improved-- technology) than competitors do.
- 3) By managing linkages among activities better than competitors do.

R&D and Competitive Advantage

On the basis of the previous discussion, the choice of which value chain activity should a firm focus on should be governed by the competitive advantage(s) that the firm is pursuing in implementing its competitive strategy. In other words, if low cost-low price is the strategy, then low-cost technologies should be used, consistent with maintaining acceptable levels of quality, availability, attractiveness, and so forth. (Obviously, technology choice interacts with other strategic variables. For example, low unit costs are often achieved through economies of scale which in the past have depended on mass-production technologies and large customer markets demanding standardized products and services.) Similarly, if differentiation is the strategy, then activities such as R&D which maximize the specific competitive advantage in terms of providing products capable of higher performance, innovative products, etc, should be used, consistent with the price premium customers are willing to pay for the uniqueness.

Activity 1

Think of an organization of your choice and apply the value chain analysis by categorizing the organization into primary and secondary support activities.

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10.7 DEVELOPMENT OF R&D STRATEGY

One standard definition of R&D states that research is an undirected basic science or a directed or applied science where development in research can result in innovative products and processes from either of these sources. Research differs from development of a product in that it is an exploration of ideas rather than a tangible output to be marketed. There is no guarantee that a new idea will be successful. R&D must provide a high degree of

insurance so that future gaps can be closed to meet corporate goals, such as profitability. Therefore, many innovations are necessary and should be allowed to flourish to assure the achievement of these goals. Invention is an idea that must be converted to practice, i.e., one must show that the technical idea is feasible and can be demonstrated. The payoff of a timely invention can make a firm gain a market edge by lowering price, even with a strategy that places it below production cost. The belief is that increased sales would ensure future cost reductions from increased volume of output due to increased sales.

The selection of R&D projects is the most important of all the R&D management activities. Unless an R&D department is working on the right R&D projects, all its efforts could come to zero. Consequently, most R&D organizations spend a great deal of time trying to select the right projects. What is rarely seen in these efforts to select the right R&D projects, however, is the value of having an overall R&D strategy before selecting R&D projects. A R&D strategy is important because it helps a R&D organization select projects in terms of a broader outlook rather than just bit by bit. When a R&D department has an overall strategy and has some strategic goals, it will more likely select projects that are in tune with its technical strengths, the capabilities of the company, and the demands of the marketplace. In order to develop a sound R&D strategy, certain prerequisites must be in place.

Pre-conditions for Effective Development of an R&D Strategy

In order to develop a R&D strategy, a R&D organization first must make sure that certain conditions are in place. There are seven **pre-requisites** for developing a R&D strategy:

- 1) A belief that a R&D strategy can solve problems.
- 2) Creation of a planning staff within a large R&D department or the establishment of a strong commitment by the R&D line managers in a small R&D organisation to devote enough effort to doing R&D strategic planning.
- 3) A method of linking R&D strategic planning to R&D operations.
- 4) A mode of getting strategic marketing done.
- 5) The active support from senior management.
- 6) Prior efforts to develop a R&D strategy.
- 7) A series of substantial and solid efforts that produce tangible results on their own, but allow R&D people to get better at R&D strategic planning.

10.8 STEPS INVOLVED IN DEVELOPING R&D STRATEGY

1) Belief that R&D Strategy solves Problems

Although the idea of developing a R&D strategy may be accepted, usually development of an R&D strategy will not come about unless a R&D strategy is used for solving a problem; which means, the effort that goes into developing a R&D and the conflicts that may occur about how R&D resources should be allocated are so great that unless

a R&D organization has a clear R&D strategy it rarely will make the effort or face the conflicts. On the whole, R&D organizations require a strategy when they wish to use new technologies for future products. If, on the other hand, a R&D organization relies solely on utilizing existing technology to develop new products, then it may not need a R&D strategy.

Usually, R&D managers recognize that a R&D strategy will help them when they have doubts about how R&D resources are being used. For instance, the R&D managers of a FMCG company perceived the need for R&D strategy when they were forced to consolidate all R&D on coffee that previously was carried out in several laboratories. To accomplish this task, they recognized that they had to have a R&D strategy to prioritise and coordinate and all R&D being done on coffee. Later they perceived the need to develop a R&D strategy related to R&D being done on beverages in a few laboratories. In short, although developing a R&D strategy is a nice idea, this idea will not be put into practice unless R&D managers perceive that a R&D strategy can solve a problem.

2) **Creating Planning Staff within a R&D Organization**

Creating a planning staff or establishing a strong commitment by the R&D managers in a small R&D organization to devote enough effort to R&D strategic planning is necessary. Although line managers within a large R&D organization by themselves can develop a R&D strategy, in practice, if there is no planning staff to facilitate the development of a R&D strategy, there will never be a R&D strategy.

Although some companies have a R&D strategy, it was not easy to develop one. The R&D planners normally run into two types of problems: analytical and organizational problems. The analytical problems surface when the R&D planners first attempt to facilitate the development of a R&D strategy. They find that there is no accepted methodology for developing a R&D strategy. Although R&D managers in some companies, consultants, and academicians all had their opinions on what a R&D strategy should be and on how it should be formulated, almost none of them have a complete picture of the process. In addition, the opinions of these various R&D managers, consultants, and academicians often were in conflict with each other or were irreconcilable. Thus, the R&D planners have to develop their own methodology. In addition, R&D planners also find that members of the R&D staff are generally not interested in developing a R&D strategy. Because of this, R&D planners have to spend majority of their time during the first few years persuading the R&D staff to do R&D strategic planning and then educating them with regard to how a R&D strategy can be developed.

The experiences of these R&D planners are typical. Someone usually has to develop the methodology to be used in doing R&D strategic planning. Someone also has to be the champion of R&D strategic planning, or it will not get done. In theory, R&D managers in a large R&D organization can handle these responsibilities. In practice, R&D

line managers in a large R&D organization normally have so many other responsibilities that they neglect R&D strategic planning. Thus, a R&D planning group usually proves to be necessary to getting R&D strategic planning done.

In a small R&D organization, on the other hand, R&D line managers are the only ones who can develop an R&D strategy because staff positions rarely exist. Thus, to get R&D strategic planning done, the R&D line managers in a small R&D organization must add the responsibilities of an R&D planning group to their normal responsibilities. These R&D managers usually will not be able to devote much time to developing a planning methodology. On the other hand, because they have responsibility for managing the R&D groups, if the R&D managers in a small R&D organization do develop an R&D strategy, they should have less difficulty in getting this strategy accepted and implemented. The key to planning in either situation is that the doers, not the planners, must do the planning. The role of the planners is to facilitate the planning process, which is an important, although seldom appreciated, role.

3) Connecting R&D Strategic Planning to R&D Operations

To get a R&D staff to develop a R&D strategy, R&D planners (or R&D line managers) have to find ways to relate R&D strategic planning to R&D operations. This connection between R&D strategic planning and R&D operations has two aspects.

Members of the R&D staff have to be able to see that their interests are served through developing a R&D strategy. Thus, R&D planners must find a mechanism that involves R&D strategic planning and at the same time the R&D staff. One of the primary purposes of such forums is to get these R&D people to talk with each other. This cross disciplinary forum can turn out to be a useful mechanism not only for improving communication, but also for getting R&D strategic planning accepted. Once involved in interdisciplinary forum, these R&D people can get interested in technical activities as well.

The R&D projects that are actually selected and the R&D strategy need to be meaningful. In other words, a R&D strategy is not meaningful if it does not influence which R&D projects are selected and carried out. A R&D organization in a household products company addressed viewing the development of a R&D strategy as involving two phases. In the first phase the senior R&D managers defined the overall direction of the R&D strategy. During the second phase, middle-level R&D managers defined the R&D strategy through the projects they selected and carried out.

4) Linking R&D Efforts/Strategy to Customers' Needs

Although it is important for a R&D organization to link its strategic operations, this is not enough. For a R&D organization's strategic plans, this strategic plan must be clearly linked to customers' future needs. Therefore, besides doing R&D strategic planning, a R&D organization must also get marketing done in its company. Two of the hard questions that must be addressed in a strategic marketing are:

- i) Who will the company's customers be in the future?
- ii) What will those customers need in the future?

In answering these questions well, one must understand economic, regulatory, geographic, and social trends and opportunities.

5) **Active Support of the Senior Management**

To get its R&D strategy integrated with business plans, an R&D organization must have the active support of senior business managers. To gain the active support of senior business managers, an R&D organization must explain the value of R&D in business terms- for example, with regard to how R&D will help the company a) satisfy customers' needs, b) cut costs, c) expand into new markets, or d) minimize distribution problems.

Those R&D organizations that have been able to gain the active support of senior business managers for their R&D strategy were helped by organizational factors. For example, in a chemical company the R&D organization was able to gain the active support of senior business managers because the chief executive officer is one of the major proponents of R&D in the company. As opposed to most of the other senior business managers in this company, this chief executive previously managed the division of the company that sells to industrial customers. Because the customers of this division are more knowledgeable about what R&D can contribute, this CEO realized how valuable R&D can be and thus has not only supported R&D strategic planning but also has actually encouraged the R&D organization to do it.

- 6) In another company, a key factor that allowed the R&D organization to first gain and then maintain the support of senior business managers was leadership among senior business managers. Stability and continuity in leadership was important because it took a few years for the senior business managers of having a R&D strategy. By learning year after year what the R&D was trying to accomplish with a R&D strategy, the senior management was able to appreciate the benefits. The R&D department, in turn, could build progress that it had made with these senior business managers in previous years.

The key to taking each of the steps discussed above is picking a problem that the R&D organization is facing that also calls for a more systematic analysis of the use of R&D resources. The value of carrying out a series of concrete steps is that together they serve as stepping-stones to improve R&D strategic planning. Moreover, because they produce tangible results along the way, they also help elicit support for the R&D strategic planning process.

- 7) One way in which a R&D organization can maintain the vitality of planning process is through carrying out a series of concrete efforts that produce tangible results on their own, but also allow R&D people to improve planning expertise. A second method involves benchmarking studies in which the R&D people compare the strengths and weaknesses of their company's technologies in relation

to the strengths and weaknesses of their competitors' technologies. The R&D people can gain two things through this effort: 1) they gain a much better understanding of how their company stands in relation to competitors and 2) they learn how to analyze their technologies. For example, they learn how to think more precisely about how their technical work could improve the performance of the company's products. A third effort that the R&D organization can consider involves using the techniques of portfolio management to evaluate the potential and payoffs of various technologies.

10.9 PROGRESS OF R&D ORGANIZATIONS IN STRATEGY DEVELOPMENT

Few R&D organizations in India have a R&D strategy or have established the conditions required for developing a R&D strategy. For example, few R&D organisations have perceived a R&D strategy as a way to solve a problem. Many large R&D organisations do not have a R&D planning group, and organizations do not do R&D strategic planning. Also, few R&D organisations found a way to get strategic marketing done. Finally, in most companies R&D strategic planning does not have the active support of senior management.

In addition, if one looks closely at the strategic goals of those R&D that have them, one also finds that many of these goals were arrived analytically. That is, many of the strategic goals of such R&D organisations have are intuitively obvious. For example, most of the strategic goals that organizations have formulated are similar and quite predictable - 1) to ward out fundamental threats to the company's businesses, 2) to comply with government regulations.

Few R&D organizations have conducted benchmarking their technologies to competitors' technologies. Few R&D organizations carry out technology forecasting studies aimed at understanding 1) what technological changes may be occurring in the next 5 to 10 years and 2) what those technological changes may mean. Even those R&D organizations that have made significant progress in an R&D strategy admit that they still have much to do to improve the planning process. For example, the R&D organization may have a R&D strategy, but the top management does not accept this R&D strategy. Many R&D departments develop a R&D strategy, but they do not find a way to integrate their R&D strategy with the marketing and manufacturing strategies in the companies. In addition, the R&D strategy can be altered at the whim of anyone who at a later date becomes involved in the planning process without necessarily informing any one else about the changes in the plans.

In summary, some R&D organizations have made significant progress in developing an R&D strategy. Most R&D organizations, however, have barely started developing an R&D strategy. The immediate challenge facing most R&D organizations, therefore, has to do with first establishing the pre-conditions required for developing an R&D strategy. After doing this, they will then be able to meaningfully deal with issues as to what the R&D priorities should be and how R&D resources should be allocated.

Give examples of any five firms (in the Indian context), which have adopted R&D strategy as a part of their competitive strategy.

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10.10 SUMMARY

An essential component of competitive strategy is recognizing the role that R&D plays in the competitive success of a firm, and acting to ensure that technology decisions and policies contribute to the firm's competitive advantage. A firm, which has a strong R&D programme, can influence all these factors positively and contribute to the competitive advantage of a firm. Competitive advantage views R&D activities as a way of improving a process, thus reducing costs, or providing customers with the best-of-class benefit.

The selection of R&D projects is the most important of all the R&D management activities. Unless an R&D department is working on the right R&D projects, all its efforts could come to naught. A R&D strategy is important because it helps an R&D organization select projects in terms of a broader outlook rather than just bit by bit. When a R&D department has an overall strategy and has some strategic goals, it will more likely select projects that are in tune with its technical strengths, the capabilities of the company, and the demands of the marketplace. In order to develop a R&D strategy, certain pre-requisites must be in place such as belief in R&D strategy as a tool to solve problems, establishing a dedicated strategic planning group within the R&D department and the top management's commitment to the R&D strategy.

Some companies in India have made significant progress in developing a R&D strategy. Most companies, however, have barely started developing a R&D strategy. The immediate challenge facing most firms, therefore, has to do with first establishing the pre-conditions required for developing an R&D strategy. After doing this, they will then be able to meaningfully deal with issues as to what the R&D priorities should be and how R&D resources should be allocated.

10.11 KEY WORDS

Value Chain: It is a systematic way of examining all the activities of a firm performs and how they interact for analysing the sources of competitive advantage.

Competitive Advantage: Competitive advantage refers to factors that allow a company to produce goods or services better or more cheaply than its rivals.

R&D Strategy: Includes activities that companies undertake to innovate and introduce new products and services. It is often the first stage in the development process.

10.12 SELF-ASSESSMENT QUESTIONS

1. Explain the importance of R&D strategy in enhancing the competitiveness of a firm.
2. Describe the features of technology package.
3. How does R&D Strategy support the Porter's generic competitive strategies?
4. Discuss the various steps involved in development of R&D strategy.
5. Give reasons for R&D strategies not taking root in the Indian context.

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UNIT 11 KNOWLEDGE MANAGEMENT (KM)

Objectives

After reading this unit, you should be able to:

- explain the concept of Knowledge Management (KM);
- acquaint yourself with the sources and types of knowledge;
- provide a framework for developing KM systems; and
- discuss the benefits and challenges of implementing KM.

Structure

- 11.1 Introduction
- 11.2 Knowledge and Knowledge Management
- 11.3 Sources of Knowledge
- 11.4 Knowledge Creation
- 11.5 Knowledge Management Framework
- 11.6 Benefits of Knowledge Management
- 11.7 Pioneers in Knowledge Management
- 11.8 KM Initiatives in Indian Organizations
- 11.9 Software for Knowledge Management
- 11.10 Trends and Challenges in Knowledge Management
- 11.11 Summary
- 11.12 Key Words
- 11.13 Self-Assessment Questions
- 11.14 References/Further Readings

11.1 INTRODUCTION

Present day's organizations, large or small, gather vast amount of knowledge during the course of their regular operations. This knowledge remains in the 'minds' of organizational members conducting the operations- be it research, design, development, manufacturing, or services. This wealth of organizational knowledge (generally referred to as knowledge assets, knowledge capital, intellectual capital, etc.) disappears, when these persons leave the organization. The effects of losing organizational knowledge is especially noticeable and can have far reaching implications when people occupying important positions such as Chief of Designs or Chief of Marketing leave since all the knowledge they gathered over long years of experience is lost permanently to the organization, along with them. A significant amount of time must be invested in relearning and reinventing the work processes. Most organizations do have some manual system of compiling knowledge to create a corporate memory but they have not been effective mainly due to difficulties in careful organization and accessing of

the compiled knowledge. Computers have not played a significant role in this area, till recently, having mainly concentrated in the data processing field: obviously due to the faster pay back period for the money invested. However, things are changing and this field is receiving due attention from developers and organizations.

11.2 KNOWLEDGE AND KNOWLEDGE MANAGEMENT

According to Peter F. Drucker in *The New Realities*, “Knowledge is information that changes something or somebody - either by becoming grounds for actions, or by making an individual (or an institution) capable of different or more effective action”. Knowledge is viewed as a collection of facts, ideas, leanings and policies, practices and a lot more. Knowledge can also be defined more narrowly as agreed- upon explicit or formal facts, rules, policies and procedures, whereas skills are seen as information processing competencies that can generate explicit knowledge. Skills are learned by doing; knowledge is learned by studying or investigating. Knowledge can be classified as “tacit” knowledge that is personal (i.e. knowledge that cannot be expressed and communicated) and “explicit” knowledge that can be codified and expressed in a human or formal language.

Although the terms “information” and “knowledge” are often used interchangeably, there is a clear distinction between information and knowledge. Information is a flow of messages, while knowledge is created by that very flow of information and is anchored in the beliefs and commitment of its holder. Traditional management models focus on how to control the information flow and information processing within the organization. This view, however, fails to capture the essence of organization as knowledge-creating entity. What “knowledge management” should achieve is not a static management of information or existing knowledge, but a dynamic management of the process of creating knowledge out of knowledge. Hence one can argue that organizational knowledge creation is a continuous self-transcending process, which requires a new kind of management that goes beyond the traditional models of “management”.

Knowledge management is such a preposterous, pretentious and profoundly confusing phrase that many of those who really understand KM—including some of the field’s pioneers-refuse to use the term. If there is anything that those experts do agree on, it is that knowledge management is not about managing people in any traditional sense. Nor is knowledge management really about managing knowledge. They prefer terms such as knowledge sharing, information systems, organizational learning, intellectual asset management, performance enhancement, etc.

Knowledge management refers to strategies and structures for maximizing the return on intellectual and information resources. Because intellectual capital resides both in tacit form (human education, experience and expertise) and explicit form (documents and data). KM depends on both cultural and technological processes of creation, collection, sharing, recombination and reuse. The goal is to create new value by improving the efficiency

and effectiveness of individual and collaborative knowledge work while increasing innovation and sharpening decision-making.

Knowledge Management (KM)

KM is the collection of processes that govern the creation, dissemination and utilization of knowledge. In one form or another, knowledge management has been around for a very long time. Practitioners have included philosophers, priests, teachers, politicians, scribes, librarians, etc. The importance of knowledge as a key source of competitive advantage is now well established in management studies. Knowledge is undoubtedly an indispensable resource to create value for the next generation of society, industries, and companies. Yet, despite all the discussions and attentions in both the academic and business worlds, very few have articulated how organizations actually create and manage knowledge. Many companies still seem to remain locked in the phase of building efficient and effective information technology (IT) systems when they try to “manage knowledge”.

Ultimately, knowledge management is really just a way of looking at the world of business. It's a realization that who and what are assets of the organization. And just like building, operating and managing physical assets, knowledge assets need to be managed for the greatest possible return on investment.

Knowledge Management (KM) is a management technique to effectively manage knowledge in organization. It comprises of:

- Construction (compilation/collection /collation) of knowledge generated within the organization (and from external sources) into a Knowledge Repository (KR).
- Organize the knowledge so constructed in the organization in such a way that it is easily accessible to all those in the organization, who need it, on time.
- A management system that maintains the KR up-to-date and relevant.
- A set of tools, including computer systems that facilitates ease of accessing the KR as well as building and maintaining it.

A Few Facts of Knowledge Management

KM is not Useless

The entire idea sits on the fact that it's a long-term strategy to maintain the existing knowledge of the person/organization and also to harvest the “new” knowledge, which a person acquires during his process of learning. Debating what knowledge management “is not” is pointless. People intuitively know whether they are managing their own knowledge well and whether their organization helps them to work without stress and inefficiency.

People and Technology

People should not have to choose between knowing a little about a lot or about a little. They should be able to concentrate most on what they need to know most and, when needed, find out a lot about related things. This requires a browseable knowledge environment designed the way people think. People want to solve problems, think, and collaborate. They do not want to “use technology”. Technology is a means, not an end. Technology must serve people, not the other way around. Technology creates knowledge

management problems faster than it creates knowledge management solutions. If knowledge management were intuitive, organization would have perfected it by now.

Financial Factor

The cost of not managing knowledge greatly exceeds the cost of managing important knowledge. Organizations have the habit of externalizing the cost of not managing the knowledge to their customers.

Future Trends

Those who want to think and act in integrated, creative ways and solve complex problems need rich, integrated, up-to-date knowledge management environments to support them. The gulf between traditional and knowledge-driven organizations is growing as knowledge-driven organizations concentrate not only on present success but their own evolution so they can better take advantage of the new knowledge-intense environment.

The Paradoxical Image

Call it the knowledge management paradox: those who are so busy “putting out fires” that they have no time to tackle knowledge management are those who most need to manage their knowledge better. While many CEOs put KM as the top priority, few companies are still at a stage of implementation: It’s the mind shift of the organizational heads to add knowledge to the balance sheet. What we know now is that, those companies that crack strategic knowledge management will be those most likely to succeed in the new economy. The new economy is always termed as the knowledge economy. Hence a company with higher knowledge quotient makes it big!

11.3 SOURCES OF KNOWLEDGE

The **two** sources of Knowledge are:

Internal sources emerging from the operations of the organization- internal sources include the organizational operations such as design, development, engineering, sales, marketing, manufacturing, customer contact, etc. This is the basic source of organizational information, which is controllable and can be easily canalized to KR. In the absence of any formal mechanism, this knowledge remains in the minds of organization members and usually, disappears with them.

External sources such as Industry/Professional Associations, Commercial web sites, etc. - there are many professional bodies such as IEEE, academic bodies such as universities, research institutions, industry associations such as NEMA, and commercial organizations. These sources usually make the knowledge available through web sites and some times through publications. Some of them could be free services and some could be for a fee. A well-designed KM should be able to take advantage of both the sources to create and maintain a KR and allow members to easily access the knowledge stored inside it.

Types of Knowledge

There are two kinds of knowledge- **explicit** knowledge and **tacit** knowledge.

Explicit knowledge can be expressed in words and numbers and shared in the form of data, scientific formulae, specifications, manuals and the like. This kind of knowledge can be readily transmitted across individuals formally and systematically. **Tacit knowledge**, on the other hand, is highly personal and hard to formalize, making it difficult to communicate or share with others. Subjective insights, intuitions, and hunches fall into this category of knowledge. Difficult to verbalize, such tacit knowledge is deeply rooted in an individual's actions and experience, as well as in the ideals, values, or emotions he or she embraces.

These two types of knowledge are complementary to each other, and both are crucial to knowledge creation. They interact with and change into each other in the creative activities of human beings. Understanding this reciprocal relationship between explicit knowledge and tacit knowledge is the key to understanding the knowledge-creating process. The interaction between the two types of knowledge can also be called as the knowledge conversion. Knowledge is created through such interactions among individuals with different types and contents of knowledge.

Knowledge creation in organizations takes place primarily through the dynamic process of four different modes of conversion between the two dimensions of knowledge.

Socialization: Tacit knowledge to conversion takes place when tacit knowledge within one individual is shared by another through training.

Combination: Explicit knowledge to explicit knowledge conversion takes place when an individual combines discrete pieces of explicit knowledge into a whole new concept.

Externalization: Tacit knowledge to explicit knowledge conversion can be said to take place when an individual is able to articulate the foundations of his and her tacit knowledge.

Internalization: Explicit knowledge to tacit knowledge conversion takes place when new explicit knowledge is shared throughout the firm and other members begin to use it to broaden, extend and reframe their own tacit knowledge.

The Factors that Constitute the Knowledge-conversion Process:

Socialization: From Tacit to Tacit	
Tacit knowledge accumulation	Managers gather information from sales and production sites, share experiences with suppliers and customers and engage in dialogue with competitors.
Extra-firm social information collection (wandering outside)	Managers engage in bodily experience through management by wandering about and get ideas for corporate strategy from daily social life, interaction with external experts, and informal meetings with competitors.

Strategic Enablers

Intra-firm social information collection (wandering inside)	Managers find new strategies and market opportunities by wandering inside the firm.
Transfer of tacit knowledge.	Managers create a work environment that allows peers to understand craftsmanship and expertise through practice and demonstrations by the master.
Externalization: From Tacit to Explicit (Creating Concepts)	
	Managers perform facilitation of creative and essential dialogue, the use of “adductive thinking”, the use of metaphors in dialogue for concept creation.
Combination: From Explicit to Explicit	
Acquisition and integration	Managers engage in planning strategies and operations, assembling internal and external existing data by using published literature, computer simulation and forecasting.
Synthesis and processing	Managers build and create manuals, documents, and databases on products and services and build up material by gathering management figures and /or technical information from all over the company.
Dissemination	Managers engage in planning and in implementation of presentations to transmit newly created concepts.
Internalization: From Explicit to Tacit	
Personal experience: real-world knowledge acquisition	Managers engage in “enactive liaisoning” activities with functional dept. by using cross functional development teams. Search and share new values and thoughts: share and try to understand management visions and values through communications with fellow members in the organization.

Stimulation and experimentation; virtual-world knowledge acquisition	Managers engage in facilitating prototyping and benchmarking and facilitate the challenging spirit within the organization; managers form teams as a model and conduct experiments and share results with the entire dept.
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Source: Adapted from Nonaka, Byosiene, Borucki, and Konno (1994).

Activity 1

Give examples of any three organizations in the Indian context which, in recent times have become success stories as a result of knowledge conversion.

- a)
- b)
- c)

11.4 KNOWLEDGE CREATION

Most, if not all, firms in developing countries are engrossed in activities to catch up with advanced countries. Even the majority of firms in advanced countries are engaged in catching up, as not all firms can be pioneers of novel breakthroughs even in these countries. Nonetheless, research on organizational knowledge creation and innovation is concentrated mainly in advanced countries and is focused mostly on the pioneering process. Research on those subjects in the catching up process, particularly in developing countries, is, however, scanty.

Knowledge creation, whether for imitation or innovation, takes place at two levels- individual and organization. The prime actors in the process of knowledge creation are individuals within the organization. Knowledge creation in organizations is not, however, the simple sum of knowledge creation by individuals. Rather it is the process that creates knowledge, which is distributed across the organization, is communicated among its members, has consensual validity, and is integrated into the strategy and management of the organization. Individual knowledge creation is, therefore, an indispensable condition for knowledge creation in the organization but cannot be the sufficient one. Organizations create knowledge only when individual insights and skills become embodied in organization routines, practices, and beliefs. Only effective organizations can translate individual knowledge creation into organizational knowledge creation.

Knowledge creation is a function of an organization's absorptive capacity. Absorptive capacity requires a learning capability and develops problem-solving skills. Learning capability involves the development of capacity to assimilate existing knowledge (for imitation). While problem-solving skills represent a capacity to create new knowledge (for innovation).

Absorptive capacity has two important elements: prior knowledge base and intensity of effort. **First**, the prior knowledge base refers to existing individual units of knowledge available within the organization. Accumulated prior

knowledge increases the ability both to make sense of and to assimilate and use new knowledge. Relevant prior knowledge base includes basic skills and general knowledge in the case of developing countries, but it includes the most recent scientific and technological knowledge in the case of industrially advanced countries. Thus prior knowledge base should be assessed in its relation to the task difficulty involved. **Second**, the intensity of effort refers to the amount of energy expended by organizational members to solve problems. Learning how to solve problems is usually built up over many practice trials involving related problems. Such effort intensifies interaction among organizational members that facilitates knowledge conversion and creation at the organizational level.

11.5 KNOWLEDGE MANAGEMENT FRAMEWORK

All mid –sized and bigger organizations need to look at KM seriously- It is not just a fad anymore nor a far of concept- it is real and here and is now available- it is of strategic importance and we cannot afford to postpone it any longer except at the cost of survival. Each industry has its own KM system evolved to suit its needs. Though the entire process starts with a bang, but somewhere down the line the entire process stands still. Following on the CMM Working model, we can devise a similar model for effective working of a KMS in the organization as depicted below in figure 11.1.

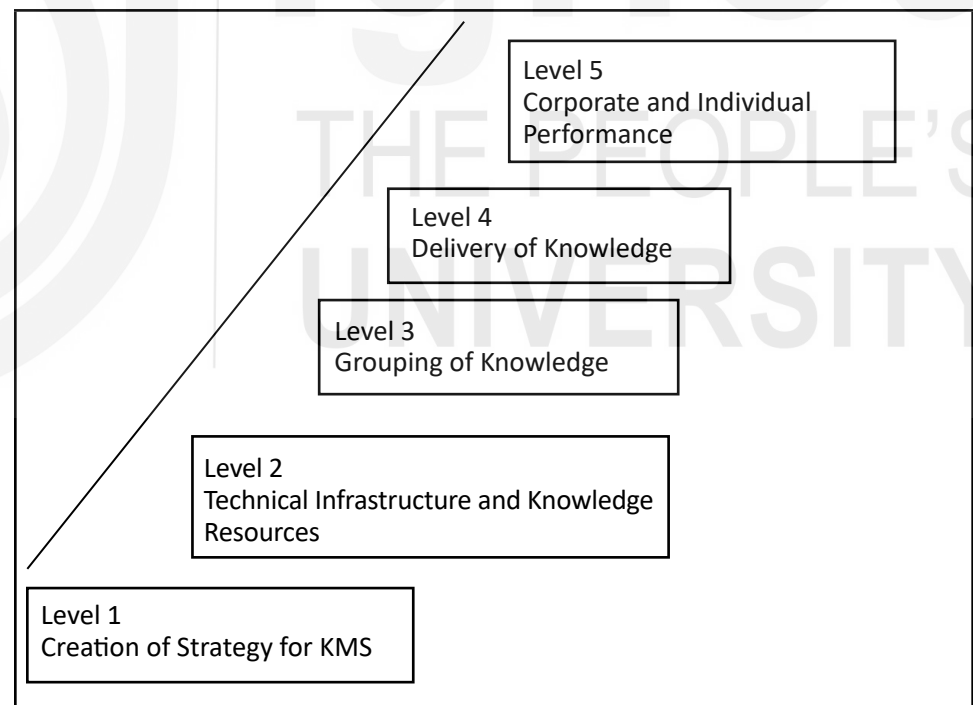


Figure 11.1: Working of a KMS in an Organization.

Level 1: Creation of a Strategy for KMS

This level involves a process where the Best Practices of the organization are listed and accumulated. A strategy is evolved for accumulation of entire gamut of the organizational and personal knowledge. Identify organizational needs/objectives in line with the systems needs/ objectives. Establish policies, which are to be adhered to.

Instead of putting the entire process to be unpredictable, we create a process, which makes the organization run on a particular schedule. Performance of the system is gauged by the capability of the individuals and the organization collectively with their innate skills, knowledge and motivations. Policies for managing the system, implementation procedures are established. An effective process can be characterized as practiced, documented, enforced, trained, measured, and able to improve.

Level 2: Technical Infrastructure and Knowledge Resources

This level would involve the design of repository, collaboration, dissemination, and hardware specifications. Technology requirements as in capture, store, search, retrieve, message, structure, navigate, share, collaborate, synthesize, solve, recommend, integrate, are maintained in detail in the process and functionality scope of the system. The output of one process would/may overflow as the input of another process. Hence knowledge need not necessarily be sequential. After knowledge is acquired, synthesized, or created, it needs to be codified, accessed and transferred again. The knowledge cycle is supported by collaboration.

Functional scope would include the technology, comprising of KM. Semantic functionality utilizes technologies such as clustering, categorization, linguistics analysis, data extraction, and knowledge maps. It would also include functionality filters identification of experts, and alignment of users into communities (net groups threaded conversations, network directories). Visualization addresses navigation and retrieval. Scale addresses the range of data types and physical locations available to KM users.

Technology requirement scope lists technologies that collectively comprise full function KM. Not every enterprise needs all functions, and no KM vendor/tools fulfill all of these requirements. The selection of technologies is preceded by the clear definition of KM strategy and scope. KM deals with enterprise-wide structured/unstructured data source, representing multiple data types and formats. KM must bridge knowledge resources to be accumulated as in the process of existing knowledge, knowledge sharing, knowledge discovery, identification of intellectual assets of the personnel of the organization and implementation of KM framework.

The level 2 of the system should allow us to summarize the planning and tracking of the entire process. The process is under the control of a knowledge office and each knowledge worker has the right to make the necessary contribution to the system.

Level 3: Grouping of Knowledge

This level would involve the proper grouping of all the knowledge accumulated during level 2. Knowledge can be classified in the following groups: Individual Group/Project/Corporate.

Individual Knowledge: Acquisition of knowledge would involve the following, search, filter, integrate, capture, store, validate and personalize the entire process. It would also let the user learn, analyze, interpret, comment, evaluate, explore and classify the intellectual asset of the person.

Group/Project Knowledge: Individualistic knowledge to be integrated with the group. What counts here are the integration, management and recording capability of the group knowledge. Each individual should be competent enough to know what is happening in his/her group.

Corporate Knowledge: This would cover the knowledge of the organization, policies, practices, market trends, industry trends and the like, integration of external sources in the knowledge base. Monitor, comment and synchronize the entire process of the organizational knowledge. At the end of this level, all these three groups are to be in sync with each other. Attention should also to be given to peer groups, inter-group coordination, intra-group coordination, training program, organization process, and organization process focus. This level should also ensure that the process is based on a common, organization-wide understanding of the activities, roles, and responsibilities in a defined process.

Level 4: Delivery of Knowledge

The level would start with the strategy of identifying audience for all the resources accumulated. Personalization of the knowledge acquired would be done here. Audience identified would have to be summarized and the entire visual to be presented to them for proper usage to establish a common bond and provide seamless access.

Level 5: Corporate and Individual Performances

This level would require the knowledge office to measure the performances of the corporate and the individual performances after effective dissemination of knowledge through the KMS. The information, which has been gathered, has to be used effectively for better performance of the organization and the individual.

An effective KM should have the following feature: acquisition, creation, packaging, or application of internal and external knowledge as a primary objective of the work. The inputs and the outputs will have high levels of variability and to a certain extent some degree of uncertainty equivocally, and incompleteness. The process of knowledge work tends to be variable, and difficult to analyze as a set of procedures or steps. Characterized by exception rather than routine, it is performed by professional or technical workers with a high level of skill and expertise. Knowledge and work processes include such activities.

Any KM project should essentially have the following four components:

- **Create KR (Knowledge Repository):** Involves finding and collecting internal knowledge and best practices. Some of the knowledge may be found in organizational documents such as memos, reports, and presentations and can be easily stored in a repository. Other knowledge is discovered through discussion. Discussion databases are another form of knowledge repository. Tools like Lotus Notes and Microsoft Exchange Server facilitate these discussion databases.
- **Improve Knowledge Access:** Involves determining ways to facilitate finding the person with the required knowledge and then transferring the knowledge to another person. Sometimes, simply storing the knowledge in a repository is not sufficient. Face to face transfer of

knowledge can be more effective. Technology, such as desktop video conferencing, can enable face-to-face knowledge access.

- **Enhance Knowledge Environment:** Involves changing the way people work. Employees are encouraged to share knowledge as well as reuse existing knowledge. Coaching and training in learning and sharing practices will probably be necessary. This may be a very difficult task if the organization's culture does not currently share information.
- **Manage Knowledge as an Asset:** Involves demonstrating that effectively using the knowledge management repositories or face-to-face transfer of knowledge has allowed the organization to save or make money. This is currently difficult, but will probably become more accepted as knowledge management becomes more widely used.

11.6 BENEFITS OF KNOWLEDGE MANAGEMENT

Some of the tangible benefits of knowledge management are directly related to their bottom line savings. In today's information-driven economy, companies continuously tap most of the opportunities and ultimately derive most value from intellectual rather than physical assets. According to many experts, to get the most value from a company's intellectual assets, knowledge must be shared and served as the foundation for collaboration. Consequently, an effective KM program should help a company leverage the assets and provide the following benefits:

- Fostering innovation by encouraging free flow of ideas.
- Improving customer service by streamlining response time.
- Boosting revenues by getting products and services to market faster.
- Enhancing employee retention rates by recognizing the value of employees' knowledge and rewarding them for it.
- Streamlining operations and reducing costs by eliminating redundant or unnecessary processes.
- A creative approach to KM can result in improved efficiency, higher productivity and increased revenues in practically any business function.

Knowledge Management's (KM) role is to connect knowledge owners with knowledge seekers. The knowledge of one is transferred to the mind of the other, so that a new decision can be made or situation can be handled. KM provides a means to capture and store passing knowledge and broker it to the appropriate individuals.

Examples where Knowledge Management can offer greatest benefit

KM is beneficial especially in both social and business segments. Here are a few examples where KM could provide great benefits:

Social (Governmental)

1. Coping with natural disasters
2. Safety in aviation/railways
3. Research and education

Business

1. **Research and Development:** Faster solutions and reduction of redundant research.
2. **Design and Development:** By allowing easier access to past design documents, best practices and better designs become available, faster, making faster deliveries gaining a competitive edge for the organization.
3. **Operations:** Problems reported from the field greatly aid in improving manufacturing practices and add to product quality.
4. **Cultural Change:** From striving to create and own ideas and objects- to sharing improving and reusing, in addition to create new ideas and objects.

Activity 2

Give real world examples in the following areas where KM has been beneficial.

	Company	Product
R&D		
Design & Development		
Cultural Change		

11.7 PIONEERS IN KNOWLEDGE MANAGEMENT

- 1) **Anderson Consulting:** Created a system called knowledge Exchange to facilitate the sharing of problem resolutions and project documents.
- 2) **USAA Insurance Firm:** Created a customer feedback system which increased customer loyalty, cut marketing costs, in addition to raising profit.
- 3) **GE:** GE has since 1982 collected all customer complaints in a database that supports telephone operators in answering customer calls. GE has programmed 1.5 million potential problems and their solutions into its system.
- 4) **Ritz Carlton:** All staff members are required to fill in cards with information from every personal encounter with a guest. These data plus all guest requirements are stored and printed out to all staff when the guest arrives again, so that each guest receives a personal treatment.
- 5) **Agro Corp USA:** Sells fertilizers and seed. Data on farmers' soils are combined with weather forecasts and information crops. Analyses are fed back to the farmer via sales representatives to help farmer select best combinations of crops.
- 6) **Dow Chemical USA:** Has put all its 25000 patents into a database, which is used by all divisions to explore how existing patents can gain more revenues. The experience from this application is now being transferred into other intellectual assets, like brands.

- 7) **Boeing 777 USA:** First “paperless” development of aircraft included customers in constructed sub parts, rather than usual organization design team, construction team. Suppliers worldwide used same digital databases as Boeing.
- 8) **Beckman Labs USA:** A biotech firm has reorganized itself to optimize knowledge sharing. Has created Knowledge Transfer Department to co-ordinate efforts. Employees best at knowledge sharing gain both financial rewards and management positions.
- 9) **McKinsey and Bain and Co:** These two management-consulting firms have development “knowledge databases” that contain experiences from every assignment including names of team members and client reactions. Each team must appoint a “historian” to document the work.

There are many others in the field now in India, all the leading software development organizations have some kind of KR. Especially those organizations that have been assessed at level 5 of CMM model of SEI have initiated KM activity in their organizations. Some of those KM initiatives are discussed in the following section.

11.8 KM INITIATIVES IN INDIAN ORGANIZATIONS

Knowledge Management Initiatives at Infosys

Infosys provides consultancy and software services worldwide to many companies. It employs large number of employees and executes several software projects at any given point of time in diverse areas such as telecom, manufacturing, insurance, finance, etc. “Powered by intellect and driven by values”, KM at infosys is founded on ‘Learn once and Use anywhere concept. Needless to say, managing knowledge at infosys is a huge challenge. Infosys manages organization-wide knowledge using three centrally operated systems- Knowledge Shop (K-Shop); Process Assets Database (PAD) and People Knowledge Map (PKM).

Knowledge Shop

Infosys built the K-shop architecture on Microsoft site serve technology, and all employees can access it through a web interface. The company encourages people to submit papers related to technology, domain, trends, culture, project experiences, internal or external literature, etc. They can submit the articles in any format that the web supports, and designed templates for various content types to ensure uniformity. In addition, the K-shop has an excellent search facility that offers search through multiple parameters. K-shop documents are available to all infosys employees and are segregated based on the users selected keywords and content type.

Process Assets Database

Process assets database is a database which capture the “as is” projects deliverable. This database contains the employee’s experiences on projects, projects artifacts such as project plans, design documents, and test plans. Users can search the documents, based on domain, technology, project type,

project code, customer name, and so on. This helps provide new project with information on similar, previously executed projects and helps set quantitative goals.

People Knowledge Map

The People Knowledge Map is a directory of experts in various fields. It is an internet-based system where employees can search and locate experts. It serves as the bridge between knowledge workers: the user and the provider. Infosys intranet portal SPARSH serves as the window for all systems and acts as the central tool. The company's quality system documentation is a repository of all process-related guidelines, checklist, and templates. These serve to standardize the projects' outputs. Infosys also has electronic bulletin boards for discussing technical and domain-related topics. In addition, there are news groups and newsletters brought out by various departments that discuss technology and business trends.

The distinguishing feature, however, is KM with project level focus. Ready reaction to customer request, improved productivity through rework, and teamwork are some of the benefits of this approach. Dynamic KM, which takes the form of web sites to manage knowledge content, training plan with material to tackle project attrition, weekly knowledge sharing sessions, defining the KM activities in the project plan (2%-3%) etc. - are some project related KM activities.

Incentive for knowledge sharing is another feature of KM at infosys. When an infosys submits a document to the K-shop, experts review the document in detail. If found acceptable, the K-shop publishes it. The reviewer and author are rewarded with Knowledge Currency Units (KCU) when an employee reads or uses a document for the K-shop, he or she is encouraged to give KCU, for that document based on the benefits gained from reading it. Authors can accumulate KCUs for their documents and redeem them for cash or other gifts. Thus, KCU serve twin objectives: they act as a mechanism both for rewarding knowledge sharing and rating the quality of assets in the repository.

Infrastructure supporting KM includes LAN/WAN/RAS facilities, E-mail, internet, intranet, and extranets. Voice/Fax Networking/Chat, Video Conferencing, Real Audio/Video/Web-cast, Business Solutions Hosting, Systems Management, Technology Enabled Help-desk, Customer Connectivity, etc. Recognizing and Rewarding Innovation and KM is done through funding for presentation at conferences, Publications on web sites, "Ideas beget wealth"- the syslabs awards for technical innovation, best practice sharing, KCU- the Knowledge Currency Unit, innovation is a criterion for the infosys Excellence Award, "Innovation management"- a key result area in the infosys KMM (knowledge management maturity) model.

The K-shop owns around 12,600 documents. Knowledge Management has helped infosys increase its productivity and reduce defect levels. A rough estimate shows that infosys reduced its defects level by much as 40%, thus significantly reducing the associated rework and the cost of detecting and preventing defects. Also, effective reuse has increased productivity by 3%. All of this has been possible due to faster access to accurate information

and reuse of knowledge. A team of eight full time employees designated as 'Brand Managers' help build and maintain the KM infrastructure in infosys.

Knowledge Management (KM)

Knowledge Management Initiatives at BaaN

BaaN is a world leader in powerful, innovative, easy-to-use business software. They are at the cutting edge of business technology used by industry leaders all over the world- promoting collaboration between customers and suppliers, linking people and processes across the world, and using the internet to make business faster and more cost-effective.

They have turned towards knowledge management, in keeping with the demands of time. Two departments, namely, "Knowledge Transfer" and 'Knowledge Development' are projected for this purpose. Their main objective is to empower the members with skills necessary to meet the external world. They have a centralized database system and it is christened as "SCOPUS". Intranet facility is provided for the members with their identity and password to use the system.

One of the features of BaaN is the encouragement provided to the employees for Knowledge management. "ASK HR" is one such technique that provides a chance to the employees to make use of public folders and register their doubts and genuine problems. Longer duration training programmes are provided for new recruits. The others receive short or mini programmes to update their knowledge.

BaaN's attempts to multiply knowledge could be seen in the well-maintained library for the purpose. They contain technical as well as non-technical printed material and is used by those employees who crave for knowledge.

"SPANDANA" known as 'reaction' is keenly felt in their monthly meetings. The people talk and they talk openly and freely with the management. They are helpful in extending the sharing of the knowledge, which is considered as rich source of knowledge. The meetings also make the people to come out of their shell and express their genuine concern for aspects that the organization stands for. Sharing of knowledge, beyond doubt highlights the brighter side of the employees' vast experiences in particular fields, their updated knowledge, their concern for the system and their sense of responsibility. Periodic Seminars and discussions help both in documentation and multiplying the knowledge thus leading to an effective knowledge management.

11.9 SOFTWARE FOR KNOWLEDGE MANAGEMENT

KM suites provide solutions for creating centralized repositories for storing and sharing data (knowledge) as well as providing solutions and tools for searching, retrieving and managing this data.

1. **Retrieval ware by Convera- www.convera.com:** Retrieval ware creates a complete inventory of all enterprise assets, and then enables users to search more than 200 document types on file servers, in groupware systems, relational databases, document management systems, web servers and more while respecting access rights- all from a common user interface.

2. **Rightnow Technologies- www.rightnow.com:** Knowledge Management Software. The self-learning-knowledge base completely integrates all communication channels resulting in customers and agents accessing information -whether they use self-service, submit an email or initiate a live chat session.
3. **IBM-Lotus- Knowledge Discovery Products:** A discipline providing the strategy process, and technology to share and leverage information and expertise to do just that knowledge discovery products and solutions can help rapidly achieve KM by capturing and organizing knowledge in the form of content, expertise and communication so it can be easily managed, located, evaluated and reused to drive responsiveness, innovation, efficiency and learning. This is shown in figure 11.2.

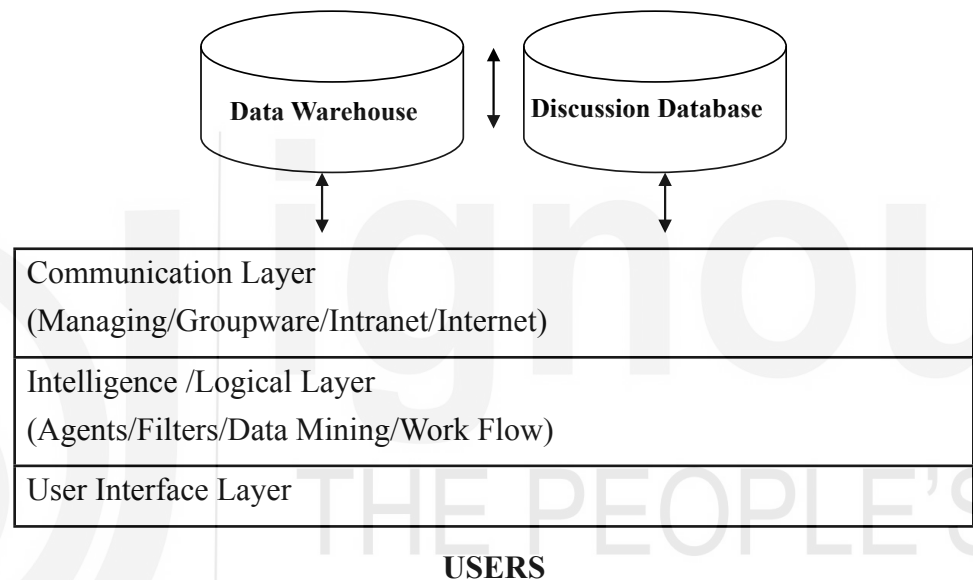


Figure 11.2: Technology Architecture

11.10 TRENDS AND CHALLENGES IN KNOWLEDGE MANAGEMENT

Many software development organizations have been assessed for SEI's Capability Maturity Model (CMM) in India and many had been assessed at level 5. This mandates that the organization have a KR- and thus most of these organizations have instituted a KR in their respective organizations. These are not accessible for public but are accessible to their organizational members.

Presently the emphasis is being focused on Information Technology related KM-creating tools and techniques that facilitate creation of KR and effectively utilizing it.

Groupware (IBM-Lotus Notes, Microsoft Exchange Servers-Outlook) has become the more popular tool. Reasons are not hard to guess-

- It facilitates Discussion Groups very effectively
- It is very economical and cost effective
- Familiarity- they are the most popular e-mail tools

- Easy to use for creation of KR and using it
- Robust and reliable tools
- However, analytical tools are absent

Data warehouse aids creation of KR. Data Warehousing has grown from data repository to Knowledge Repository. Tools like, OLAP and data mining techniques facilitate effective utilization of KR. These facilitate not only extraction of information but also analysis. However, the costs are high in this segment.

ETL (Extraction Transformation and Loading) tools facilitate extraction of information from external sources including web pages, transforming them to desired format and loading them into the organizational KR. Business intelligence (BI) tools facilitate interpretation and form inferences, BI coupled with KM is fast emerging as a very powerful Decision Support System (DSS). This trend may, in near future, provide the success and impetus to DSS, which it has been lacking so far.

Issues and Challenges in KM

“Organizations have dispersed for real and critical reasons,” says Chad Weinstein, Director of Knowledge Management consultant for Sopheon PLC, a Minneapolis-based professional services and software company. ‘It lets them get the best talent, the best resources and close proximity to potential clients or crucial suppliers.’ Along with the opportunities that come with dispersion, a globetrotting, telecommuting workforce presents challenges in managing and disseminating a corporation’s collective knowledge.

The first of those challenges is merely getting individuals within the company to communicate with each other wherever they are located, according to Daniel Rasmus, vice president and KM research leader for Giga Information Group Inc. in Aliso Viejo, Calif. “Ridiculous as it may sound, many organizations have trouble getting people to share information who aren’t on the same floor, so adding remote workers or those in other geographical locations can prove difficult,” he says.

Enterprises are realizing how important it is to “know what they know” and be able to make maximum use of the knowledge. This knowledge resides in many different places such as: databases, knowledge bases, filing cabinets and peoples’ heads and are distributed right across the enterprise. All too often one part of an enterprise repeats work of another part simply because it is enterprise impossible to keep track of, and make use of, knowledge in other parts. At Tata Steel, one incident more than any other drove home this point. In 1999, a foreign technical consultant was summoned to the Indian steel giant to solve a problem. He replied that he had already been engaged and solved it the year before. In other words, the company, despite having a sophisticated IT infrastructure, did not seem to systematically “know” what its problems were and how it had been solving them.

Enterprises need to know:

- What their knowledge assets are?
- How to manage and make use of these assets to get maximum return?

Knowledge assets are the knowledge regarding markets, products, technologies and organizations, that a business owns or needs to own and which enable its business processes to generate profits, add value, etc. Knowledge management is not only about managing these knowledge assets but also about preserving knowledge; using knowledge, and sharing knowledge. Therefore, knowledge management involves the identification and analysis of available and required knowledge assets and knowledge asset related processes, and the subsequent planning and control of actions to develop both the assets and the processes so as to fulfill organizational objectives. Success in an increasingly competitive marketplace depends critically on the quality of knowledge, which organizations apply to their key business processes. For example, the supply chain depends on knowledge of diverse areas including raw materials, planning, manufacturing and distribution. Likewise product development requires knowledge of consumer requirements, new science, new technology, marketing, etc.

The challenge of deploying the knowledge assets of an organization to create competitive advantage becomes more crucial as:

- The marketplace is increasingly competitive and the rate of innovation is rising. So that knowledge must evolve and be assimilated at an ever-faster rate.
- Corporations are organizing their businesses to be focused on creating customer value. Staff functions are being reduced, as are management structures. There is a need to replace the informal knowledge management of the staff function with formal methods in customers aligned business processes.
- Competitive pressures are reducing the size of the workforce, which holds this knowledge.
- Knowledge takes time to experience and acquire. Employees have less and less time for this.
- There are trends for employees to retire earlier and for increasing mobility, leading to loss of knowledge.
- There is a need to manage increasing complexity as small operating companies are transnational sourcing operations. A change in strategic direction may result in the loss of knowledge in a specific area. A subsequent reversal in policy may then lead to a renewed requirement for this knowledge, but the employees with that knowledge may no longer be there.

Another reason for the difference between tacit and explicit knowledge is more than academic is that, by and large, the distinction determines who owns the knowledge. Explicit knowledge is most likely the property of the firm. One way or another it is either data or work product. But since tacit knowledge cannot be codified, it effectively remains the property of the knowledge worker. Companies have certainly tried to own this knowledge. While the company employs them, knowledge workers are ethically—and sometimes contractually—prohibited from sharing their knowledge with competitors. But if the knowledge workers leave the firm they take that knowledge and its inherent value with them.

Making this distinction between knowledge and the knowledge worker makes it easier to account for knowledge assets. A knowledge worker is an asset that appreciates over time. Knowledge itself is more often a depreciating asset. Patents, for example, quickly lose their value if not licensed quickly. A sales lead becomes worthless if the contact chooses a competitor's product or leaves the customer's company for another job. Unlike other resources, however, knowledge is not subject to the law of diminishing returns; it is not depleted through use.

KM technology solutions themselves pose a challenge to implementation. Existing solutions often require customization that puts a "significant burden" on implementers. A survey with IT executives also said that they are waiting for some collaboration and communication tools to fully develop, such as voice recognition technology. The process approach allows an end-to-end view of how best to structure, sequence, and measure work activities to reach the targeted outcomes. Processes are described as "specific ordering of work activities across time and place, with a beginning, an end, and clearly identified inputs and outputs: a structure for action." The process approach promotes an examination of what and how things are done from a viewpoint of producing value for a customer. The capabilities of information technology (IT) also promote an explicit management of knowledge processes. IT can be used to capture, codify, distribute, and apply the firm's best knowledge.

Getting employees on board is a major issue in an environment where an individual's knowledge is valued and rewarded. Establishing a tradition that recognizes tacit knowledge and encourages employees to share their knowledge are ways companies motivate employees to participate in KM. Incentive programme is another way. However, there's the danger that employees will participate solely to earn incentives, without regard to the quality or relevance of the information they contribute.

KM is not a technology-based concept. It needs careful planning and analysis. While technology can support KM, it is not the be all and end all of KM. KM decisions should be based on who (people), what (knowledge) and why (business objectives). Also a KM programme should be in sync with the business goal. While sharing best practices is a good idea, there must be an underlying business reason to do so. Without a solid business case, KM is a futile exercise.

Knowledge is not static. As with many physical assets, the value of knowledge can erode over time. Since knowledge can get stale fast, the content in a KM programme should constantly updated be amended, and deleted. Further, the relevance of knowledge at any given time changes, as do the skills of employees. Therefore, there is no endpoint to a KM programme. Like product development marketing and R&D, KM is constantly evolving business practice. Another most important fact is that not all information is knowledge. Companies diligently need to be on the lookout for information explosion and overload. Here quality is the key and not quantity.

11.11 SUMMARY

The importance of knowledge as a key source of competitive advantage is now well established in management studies, as suggested by the growing literature focusing on knowledge creation and transfer. Knowledge is undoubtedly an indispensable resource to create value for the next generation of society, industries, and companies. Yet, despite all the discussions and attentions in both the academic and business worlds, very few have articulated how organizations actually create and manage knowledge. Many companies still seem to remain locked in the phase of building efficient and effective information technology (IT) systems when they try to “manage knowledge”.

It is imperative for today’s progressive organizations to integrate Knowledge Management (KM) in their routine operations. Incentive for this is the falling cost of hardware and availability of appropriate software tools to capture and manage corporate knowledge. Now the sources of knowledge are moving from paper to soft copy form and on to the web- thus- available to all, either freely or through payment of a nominal fee. This soft copy form of information/knowledge lends itself to automatic collection, transformation and manipulation besides being amenable to fast distribution to all concerned- all these can be accomplished automatically without human intervention. KM progresses through four stages namely, create Knowledge Repository, improve knowledge access, create knowledge environment and manage knowledge as an asset.

Challenges in effective implementation of KM in organization stem from new forms of acquiring information, new tools to store and manipulate information, new tools for transformation of information, new ways of working over distance and time, and to shift from information access and sharing to knowledge networking. KM has been progressing from Groupware and Data warehousing to Business Intelligence through usage of sophisticated software tools. KM can give significant benefits to organizations by assisting them to improve quality of service delivery, avert disasters and reduce costs through early recognition patterns, alerts and probable actions.

11.12 KEY WORDS

Combination: Explicit knowledge to explicit knowledge conversion takes place when an individual combines discrete pieces of explicit knowledge into a new whole.

Externalization: Tacit knowledge to explicit knowledge conversion can be said to take place when an individual is able to articulate the foundations of his and her tacit knowledge.

Explicit Knowledge: can be codified and expressed in a human or formal language. It can be expressed in words and numbers and shared in the form of data, scientific formulae, specifications, manuals and the like. This kind of knowledge can be readily transmitted across individuals formally and systematically.

Internalization: Explicit knowledge to tacit knowledge conversion takes

place when new explicit knowledge is shared throughout the firm and other members begin to use it to broaden, extend and reframe their own tacit knowledge.

Knowledge: is collection of facts, ideas, learning's and policies, practices and a lot more. Knowledge can also be defined as agreed- upon explicit or formal facts, rules, policies and procedures.

Knowledge Management: is the collection of processes that govern the creation, dissemination and utilization of knowledge.

Skills: are information processing competencies that can generate explicit knowledge. Skills are learned by doing; knowledge is learned by studying or investigating.

Socialization: Tacit knowledge to conversion takes place when tacit knowledge within one individual is shared by another through training.

Tacit Knowledge: is highly personal and hard to formalize, making it difficult to communicate or share with others.

11.13 SELF-ASSESSMENT QUESTIONS

- 1) What is Knowledge Management (KM) and how does it enhance the competitiveness of a firm?
- 2) Describe the various steps involved in a KM framework. What are the essential components of a KM project?
- 3) Search for articles and information about Indian companies that have taken KM initiatives. Highlight some important issues discussed in those articles.
- 4) What are the problems and challenges in the implementation of a KM system?

11.14 REFERENCES / FURTHER READINGS

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UNIT 12 INNOVATION

After reading this unit, you should be able to:

- acquaint yourself with the concept of innovation and creativity;
- discuss the various factors influencing innovation and creativity;
- explain the characteristics of innovative organizations; and
- familiarize yourself with the techniques for enhancing creativity.

Structure

- 1.1 Introduction
- 1.2 Concepts of Innovation and Creativity
- 1.3 Factors Influencing Creativity and Innovation
- 1.4 Characteristics of Innovative Organizations
- 1.5 The Individual and Innovation Culture
- 1.6 Fostering Creativity and the Creative Process
- 1.7 Techniques for Enhancing Creativity
- 1.8 Building Creative Organizations
- 1.9 Company Programmes to Enhance Creativity
- 1.10 Summary
- 1.11 Key Words
- 1.12 Self-Assessment Questions
- 1.13 References/Further Readings

12.1 INTRODUCTION

Post-industrial organizations today are knowledge-based organizations and their success and survival depend on creativity, innovation, discovery and inventiveness. An effective reaction to these demands leads not only to changes, in individuals and their behaviour, but also to innovative changes in organizations to ensure their existence (Read, 1996). It appears that the rate of change is accelerating rapidly as new knowledge, idea generation and global diffusion increase (Chan Kim and Mauborgne, 1999; Senge *et al.*, 1999). Creativity and innovation have a role to play in this change process for survival.

The challenge for companies is to be innovative and creative to bring to the market a stream of new and improved, added value, products and services that enable the business to achieve higher margins and thus profits to re-invest in the business. Innovation can be defined as the successful exploitation of new ideas.

12.2 CONCEPTS OF INNOVATION AND CREATIVITY

The concepts of creativity and innovation are often used interchangeably in the literature. Consequently, it is important to analyze these concepts in the context of this research. Some definitions of creativity focus on the nature of thought processes and intellectual activity used to generate new insights or solutions to problems. Other definitions focus on the personal characteristics and intellectual abilities of individuals, and still others focus on the product with regard to the different qualities and outcomes of creative attempts (Arad *et al.*, 1997; Udwadia, 1990).

Creativity as a context-specific evaluation can vary from one group, one organization and one culture to another and it can also change over time. Evaluating creativity should, therefore, be considered at the level of a person, organization, industry, profession, etc. (Ford, 1995). In this unit, the context of creativity is discussed at the level of the organization and hence, the concept of creativity can be defined as the generation of new and useful/valuable ideas for products, services, processes and procedures by individuals or groups in a specific organizational context.

Definitions of innovation found in the literature vary according to the level of analysis, which is used. The more macro the approach (e.g. social, cultural), the more varied the definitions seem to be (West and Farr, 1990). Some definitions are general and broad, while others focus on specific innovations like the implementation of an idea for a new product or service. In an organizational environment, examples of innovation are the implementation of ideas for restructuring, or saving of costs, improved communication, new technology for production processes, new organizational structures and new personnel plans or programmes (Kanter 1983 cited in West and Farr, 1990; Robbins, 1996).

West and Farr (1990) define innovation as follows: “the intentional introduction and application within a role, group or organization of ideas, processes, products or procedures, new to the relevant unit of adoption, designed to significantly benefit the individual, the group, organization or wider society”. It appears that the context in which a new idea, product, service or activity is implemented determines whether it can be regarded as an innovation within that specific context.

Innovation is often associated with change. Innovation is regarded as something new, which leads to change. However, change cannot always be regarded as innovation since it does not always involve new ideas or does not always lead to improvement in an organization (CIMA Study Text, 1996; West and Farr, 1990). An example of change that cannot be regarded as an innovation is changing office hours in an exceptionally hot summer season.

Although innovation is intangible, it is probably best described as a pervasive attitude that allows business to see beyond the present and create the future. In short, innovation is the engine of change and in today’s fiercely competitive environment resisting change is dangerous. Companies cannot protect themselves from change regardless of their excellence or the vastness

of their current resource base. Change, while it brings uncertainty and risk, also creates opportunity. The key driver of the organization's ability to change is innovation. However, simply deciding that the organization has to be innovative is not sufficient. That decision must be backed by actions that create an environment in which people are so comfortable with innovation that they create it.

Culture is a primary determinant of innovation. Possession of positive cultural characteristics provides the organization with necessary ingredients to innovate. Culture has multiple elements that can serve to enhance or inhibit the tendency to innovate. Moreover the culture of innovation needs to be matched against the appropriate organizational context. To examine culture in isolation is a mistake and to simply identify one type of culture and propose it as the panacea to an organization's lack of innovation is to compound that mistake.

12.3 FACTORS INFLUENCING CREATIVITY AND INNOVATION

Virtually all companies talk about innovation and many may actually attempt to "do it", but only a few actually succeed in doing it. The reality is that innovation, for the most part, frightens organizations because it is inevitably linked to risk. Many companies pay lip service to the power and benefits of innovation. To a large extent most remain averse to the aggressive investment and commitment that innovation demands. Instead they dabble in innovation and creativity. Even though innovation is debated in senior level meetings as being the lifeblood of the company, and occasional resources and R&D funds are thrown at it, often the commitment usually ends there. However, becoming innovative demands more than debate and resources- it requires an organizational culture that constantly guides organizational members to strive for innovation and a climate that is conducive to creativity.

The dimensions/factors of organization culture that influence creativity and innovation in an organization are discussed below. Each of these dimensions is discussed briefly to describe their influence in promoting or hindering creativity and innovation.

Strategy

An innovation strategy is a strategy that promotes the development and implementation of new products and services. The origin of creativity and innovation lies in a shared vision and mission, which are focused on the future. Furthermore, the vision and mission of a creative and innovative organization are also customer and market oriented- focusing on solving customers' problems among other things. An example of a vision that emphasizes creative and innovative behaviour is "Our company will innovate endlessly to create new and valuable products and services and to improve our methods of producing them" (Lock and Kirkpatrick, 1995).

It is also important that employees should understand the vision and mission (which support creativity and innovation) and the gap between the current situation and the vision and mission to be able to act creatively and innovatively. Having a clear corporate philosophy enables individuals to co-

ordinate their activities to achieve common purposes, even in the absence of direction from their managers. One effect of corporate statements is their influence in creating a strong culture capable of appropriately guiding behaviours and actions. However there is also a degree of doubt as to whether statements of principles have any value in driving the organization forward. Most statements encountered often are of little value because they fail to grab people's attention or motivate them to work toward a common end.

Judge *et al.* (1997) describe successful innovation as chaos within guidelines; in other words top management prescribes a set of strategic goals, but allows personnel great freedom within the context of the goals. Organizational goals and objectives reflect the priorities and values of organizations and as a result may promote or hinder innovation (Arad *et al.*, 1997). Personal and organizational goals that emphasize quality rather than effectiveness improve the levels of innovation. It appears that reflecting the value of purposefulness in the goals and objectives of organizations has an influence on creativity and innovation.

Structure

In the innovation literature, much has been written about the structural characteristics of organizations and it has been suggested that a flat structure, autonomy and work teams will promote innovation, whereas specialization, formalization, standardization and centralization will inhibit innovation. As regards the influence of organizational culture on a structure that supports creativity and innovation, values like flexibility, freedom and cooperative teamwork will promote creativity and innovation. On the other hand, values like rigidity, control, predictability, stability and order (mostly associated with hierarchical structures) will hinder creativity and innovation (Arad *et al.*, 1997).

A high level of responsibility and adaptability accompanies an organizational structure that allows for flexibility. Examples of flexibility in organizations include making use of a job rotation programmes or doing away with formal and rigid job descriptions. Freedom as a core value in stimulating creativity and innovation is manifested in autonomy, empowerment and decision-making. This implies that personnel are free to achieve their goals in an automatic and creative way within guidelines. Personnel, therefore, have the freedom to do their work and determine procedures as they see fit within the guidelines provided. Management should also believe in personnel and encourage them to be more creative by allowing them more freedom, in other words empowering them instead of controlling them (Judge *et al.*, 1997, p. 76).

Research revealed that the degree to which employees have freedom and authority to participate in decision making in solving problems determines the level of empowerment, which is positively related to the level of creativity and innovation in an organization (Arad *et al.*, 1997). The speed of decision-making can also promote or inhibit creativity and innovation. Tushman and O'Reilly (1997) claim that cultural norms that lead to quick decision-making (e.g. that speed is important and that the work rate is fast) should promote the implementation of innovation.

Co-operative teams are identified by some authors as having an influence on the degree to which creativity and innovation take place in organizations. Well-established work teams, which allow for diversity and individual talents that complement one another, should promote creativity and innovation (Arad *et al.*, 1997; Mumford *et al.*, 1997). Cross-functional teams that encourage social and technical interaction between developers and implementers can improve and promote creativity and innovation. Another important aspect is that team members should be able to trust and respect one another, understand one another's perspectives and style of functioning, solve differences of opinion, communicate effectively, be open to new ideas and question new ideas. Such effective teamwork is partly based on team members' skills and abilities and partly on the shared values within the group (e.g. values about shared trust and solving differences) (Tushman and O'Reilly, 1997).

In summary, **organic structures that promote innovation are:**

- Freedom from rules;
- Participative and informal;
- Many views aired and considered;
- Face to face communication; little red tape;
- Inter-disciplinary teams; breaking down departmental barriers;
- Emphasis on creative interaction and aims;
- Outward looking; willingness to take on external ideas;
- Flexibility with respect to changing needs;
- Non-hierarchical;
- Information flow downwards as well as upwards;
- Decision making responsibility at lower levels;
- Decentralized procedures;
- Freedom to act;
- Expectation of action;
- Belief that individual can have an impact;
- Delegation;
- Quick, flexible decision making;
- Minimize bureaucracy;

The structures that hinder innovation include:

- Rigid departmental separation and functional specialization;
- Hierarchical;
- Bureaucratic;
- Many rules and set procedures;
- Formal reporting;
- Long decision chains and slow decision making;
- Little individual freedom of action;

- Communication via the written word;
- Much information flow upwards; directives flow downwards.

Support Mechanisms

Support mechanisms should be present in an organization to create an environment that will promote creativity and innovation. Rewards and recognition and the availability of resources, namely time, information technology and creative people, are mechanisms that play this role. Behaviour that is rewarded reflects the values of an organization. If creative behaviour is rewarded, it will become the general, dominant way of behaving (Arad *et al.*, 1997). The problem is that many organizations hope that personnel will think more creatively and take risks, but they are rewarded for well-proven, trusted methods and fault-free work. Personnel should also be rewarded for risk taking, experimenting and generating ideas. Intrinsic rewards like increased autonomy and improved opportunities for personal and professional growth may support the innovation process. It is also important to reward individuals as well as teams (Tushman and O'Reilly, 1997). Management should be sensitive to which methods of reward and recognition will inspire personnel in their specific organization to be more creative and innovative (Tushman and O'Reilly, 1997).

The key attributes of organizations that reward innovation include:

- Ideas are valued;
- Attention and support of top management;
- Respect for new ideas;
- Celebration of accomplishments e.g. awards;
- Implementation of suggestions.
- Encouragement.

An organizational culture that promotes creativity and innovation should give employees time to think creatively and experiment (Shattow, 1996). In organizations where creativity and innovation are encouraged, personnel are, for example, allowed to spend 15 percent of their time on generating new ideas and working on their favourite projects. Emphasis on productivity and downsizing, which leads to more pressure on employees to work harder, is not conducive to creativity and innovation in organizations (Filipczak, 1997).

Information technology as a support mechanism is an important resource for successful innovation (Shattow, 1996). In organizations where it is part of the culture to use computer technology such as the Internet and intranet to communicate and exchange ideas, the chances of creativity and innovation taking place are improved (Bresnahan, 1997; Khalil, 1996).

Recruitment, selection and appointment and maintaining employees are an important part of promoting the culture of and specifically creativity and innovation in an organization. The values and beliefs of management are reflected in the type of people that are appointed. Apart from personality traits like intelligence, knowledge, risk taking, inquisitiveness and energy, a value like diversity is of utmost importance in the appointment of creative

and innovative people. Appointing people of diverse backgrounds should lead to richer ideas and processes that should stimulate creativity and innovation (Bresnahan, 1997; Gardenswartz and Rowe, 1998).

Behaviour that Encourages Innovation

Values and norms that encourage innovation manifest themselves in specific behavioural forms that promote or inhibit creativity and innovation. The way in which mistakes are handled in organizations will determine whether personnel feel free to act creatively and innovatively. Mistakes can be ignored, covered up, used to punish someone or perceived as a learning opportunity (Brodtrick, 1997). Tolerance of mistakes is an essential element in the development of an organizational culture that promotes creativity and innovation. Successful organizations reward success and acknowledge or celebrate failures, for example, by creating opportunities to openly discuss and learn from mistakes (Tushman and O'Reilly, 1997).

An organizational culture, in which personnel are encouraged to generate new ideas, without being harmed, and where the focus is on what is supported instead of on what is not viable, should encourage creativity and innovation (Filipczaak, 1997). Fair evaluation of ideas will also support and encourage creativity (Amabile, 1995). An organizational culture that supports a continuous learning orientation should encourage creativity and innovation. By focusing on being inquisitive, encouraging personnel to talk to one another (e.g. to clients within and outside the organization to learn from them), keeping knowledge and skills up to date and learning creative thinking skills, a learning culture can be created and maintained.

Taking risks and experimenting are behaviours that are associated with creativity and innovation. A culture in which too many management controls are applied will inhibit risk taking and consequently creativity and innovation (Judge *et al.*, 1997). The assumption that risks may be taken as long as they do not harm the organization will not encourage personnel to be creative and innovative by experimenting and taking risks (Filipczaak, 1997, p. 37). It is important that a balance should be reached in the degree to which risk taking is allowed. This can be achieved by spelling out expected results, assigning the responsibility of monitoring and measuring risk taking to someone in the organization, creating a tolerant atmosphere in which mistakes are accepted as part of taking the initiative, regarding mistakes as learning experiences, and assuming that there is a fair chance of risks being successful.

Most creative and innovative departments in an organization regard competitiveness as an important aspect of their culture. According to Read (1996, p. 226), competitiveness in organizations has shifted to the creation and assimilation of knowledge. In creating a culture of competitiveness managers should reach out to internal and external knowledge, encourage debating of ideas, create an environment in which constructive conflict will lead to information flow, support projects based on information flow and actively manage the choice of organizational design. Support for change is a value that will influence creativity and innovation positively (Arad *et al.*, 1997; Tushman and O'Reilly, 1997). Managers can create a culture that supports change by looking for new and improved ways of working, creating

a vision that emphasizes change and revealing a positive attitude towards change (Arad *et al.*, 1997; Tushman and O'Reilly, 1997). An example of a culture in which change is supported is to expect personnel, when stating their annual objectives for the year, to indicate how they intend changing their work methods.

Tolerance of conflict and handling conflict constructively are values that support creative and innovative behaviour in organizations (Mumford *et al.*, 1997; Robbins, 1997; Judge *et al.*, 1997). When there is conflict between different ideas, perceptions and ways in which information is processed and evaluated, the process of handling conflict should be handled constructively to promote creativity and innovation. Understanding different individual thinking styles and training personnel in the process of constructive confrontation will create a culture supportive of creativity and innovation.

Communication

An organizational culture that supports open and transparent communication, based on trust, will have a positive influence on promoting creativity and innovation (Barret, 1997; Robbins, 1996). Teaching personnel that disagreement is acceptable, since it offers the opportunity to expose paradoxes, conflict and dilemmas, can promote openness in communication. At the same time personnel must feel emotionally safe to be able to act creatively and innovatively and should therefore be able to trust one another, which in turn is promoted by open communication. An open-door communication policy, including open communication between individuals, teams and departments to gain new perspectives, is therefore necessary to create a culture supportive of creativity and innovation (Filipczak, 1997).

Leadership

Leading edge organizations consistently innovate, and do so with courage. It is the task of organizational leaders to provide the culture and climate that nurtures and acknowledges innovation at every level. Notwithstanding the fact that leadership is critically important, it is nevertheless insufficient on its own to build a culture of continuous improvement and innovation. To build a culture of innovation, many innovation champions must be identified, recruited, developed, trained, encouraged and acknowledged throughout the organization.

In order to build a successful and sustainable culture of innovation, leadership needs to accomplish two broad tasks. **First**, leaders need to be acutely sensitive to their environment and acutely aware of the impact that they themselves have on those around them. This sensitivity enables them to provide an important human perspective to the task at hand and is critical because it is only within this awareness that the leader can begin to bridge the gap between "leader speak" and the real world of organizational culture. The **second** factor is the ability of leaders to accept and deal with ambiguity. Innovation cannot occur without ambiguity, and organizations and individuals that are not able to tolerate ambiguity in the work place environment and relationships reproduce only routine actions. Innovative structures for example cannot have all attendant problems worked out in advance. Leaders need to build a deep appreciation of this fact; otherwise there will be a tendency to create cultures of blame. Tolerance of ambiguity

allows space for risk taking, and exploration of alternative solution spaces, which do not always produce business results. This hedges against constant deployment of tried and tested routines for all occasions. Tom Peters comes close to the mark in highlighting that most successful managers have an unusual ability to resolve paradox, to translate conflicts and tensions into excitement, high commitment and superior performance.

Characteristics that distinguish highly innovative firms against less innovative companies are as follows:

- Top management commits both financial and emotional support to innovation, and they promote innovation through champions and advocates for innovation.
- Top management has to ensure that realistic and accurate assessments of the markets are made for the planned innovation. Highly innovative firms are close to the end users, and are accurately able to assess potential demand.
- Top management ensures that innovation projects get the necessary support from all levels of the organization.
- Top management ensures that structured methodology/systems are set in place so that each innovation goes through a careful screening process prior to actual implementation.

The above suggests that senior management play a pivotal role in enhancing or hindering organizational innovation. If senior management is able to install all of the above types of procedures and practices then they effectively seed a climate conducive to innovation. It is important to note that it is not sufficient to only emphasize one or few practices.

Box 12.1: Ten Practical Steps to Keep Innovation System Alive & Well

1. **Remove fear** from your organization. Innovation means doing something new, something that may fail. If people fear failing, they will not innovate.
2. Make innovation part of the **performance review system** for everyone. Ask them what they will create or improve in the coming year and then track their progress.
3. Document an **innovation process** and make sure everyone understands it as well as his or her role in it.
4. Build in enough **looseness** into the system for people to explore new possibilities and collaborate with others inside and outside the organization.
5. Make sure that **everyone understands the corporate strategy** and that all innovation efforts are aligned with it. However, also create a process for handling the vague ideas that don't fit the strategy but are too good to throw away.
6. Teach people to **scan the environment** for new trends, technologies and changes in customer mindsets.

7. Teach people the critical importance of **diversity** of thinking styles, experience, perspectives and expertise. Expect diversity in all activities related to innovation.
8. **Good criteria** can focus ideation; however, overly restrictive criteria can stifle ideation and perpetuate assumptions and mindsets from the past. Spend the time necessary upfront to develop market and success-related parameters that will take you into the future.
9. **Innovation teams** are different from “regular” project teams. They need different tools and different mindsets. Provide enough training and coaching so that when people are working on an innovation team, they can be successful.
10. Buy or develop an **idea management system** that captures ideas in a way that encourages people to build on and evaluate new possibilities.

Source: Joyce Wycoff (<http://www.thinksmart.com>)

12.4 CHARACTERISTICS OF INNOVATIVE ORGANIZATIONS

Despite the interest in the field of innovation, much of the research evidence concerning management practices about innovation cultures and creative climate remains unsystematic and anecdotal. As mentioned earlier, the importance of culture has been emphasized by organizational theorists such as Burns and Stalker (1961), who present a case for organic structures as opposed to mechanic structures. In popular literature, Peters and Waterman (1982), similarly present arguments, which suggest that in order to facilitate innovation, work environments must be simultaneously tight and loose. Burlgeman and Sayles (1986) highlight the dependency of innovation with the development and maintenance of an appropriate context within which innovation can occur. Judge *et al.* (1997) in presenting findings from a study of R&D units compare cultures and climates between innovative and less-innovative firms and argue that the key distinguishing factor between innovative and less innovative firms is the ability of management to create a sense of community in the workplace. Highly innovative companies behave as focused communities whereas less innovative companies units behave more like traditional bureaucratic departments. They suggest four managerial practices that influence the making of such goal-directed communities.

Balanced Autonomy

Autonomy is defined as having control over means as well as the ends of one's work. This concept appears to be one of central importance. There are two types of autonomy:

- **Strategic Autonomy:** the freedom to set one's own agenda;
- **Operational Autonomy:** the freedom to attack a problem, once it has been set by the organization, in ways that are determined by the individual self.

Operational autonomy encourages a sense of the individual and promotes entrepreneurial spirit, whereas strategic autonomy is more to do with the level of alignment with organizational goals. It appears that firms that are most innovative emphasize operational autonomy but retain strategic autonomy for top management. Top management appears to specify ultimate goals to be attained but thereafter provide freedom to allow individuals to be creative in the ways they achieve goals. Giving strategic autonomy, in the sense of allowing individuals a large degree of freedom to determine their destiny, ultimately leads to less innovation. The results of strategic autonomy are an absence of guidelines and focus in effort. In contrast, having too little operational autonomy also has the effect of creating imbalance. Here the roadmaps become too rigidly specified, and control drives out innovative flair, leading eventually to bureaucratic atmospheres. What works best is a balance between operational and strategic autonomy.

Personalized Recognition

Rewarding individuals for their contribution to the organization is widely used by corporations. However, while recognition can take many forms there is a common distinction: rewards can be either extrinsic or intrinsic. Extrinsic rewards are things such as pay increases, bonuses and shares and stock options. Intrinsic rewards are those that are based on internal feelings of accomplishment by the recipient. For example, being personally thanked by the CEO, or being recognized by the peer group, being awarded an award or trophy.

Innovative companies appear to rely heavily on personalized intrinsic awards, both for individuals as well as groups. Less innovative companies tend to place almost exclusive emphasis on extrinsic awards. It appears that when individuals are motivated more by intrinsic desires than extrinsic desires then there is greater creative thought and action. Nevertheless, it has to be stated that extrinsic rewards have to be present at a base level in order to ensure that individuals are at least comfortable with their salary. Beyond the base salary thresholds it appears that innovation is primarily driven by self-esteem level rather than external monetary rewards. It appears that extrinsic rewards often yield only temporary compliance. Extrinsic rewards promote competitive behaviours, which disrupt workplace relationships, inhibit openness and learning, discourage risk-taking, and can effectively undermine interest in work itself. When extrinsic rewards are used, individuals tend to channel their energies in trying to get the extrinsic reward rather than unleash their creative potential.

Integrated Socio-Technical System

Highly innovative companies appear to place equal emphasis on the technical side as well as the social side of the organization. In other words, they look to nurture not only technical abilities and expertise but also promote a sense of sharing and togetherness. Fostering group cohesiveness requires paying attention to the recruitment process to ensure social “fit” beyond technical expertise, and also about carefully integrating new individuals through a well-designed socialization programme. Less innovative firms on the other hand appear to be more concerned with explicit, aggressive individual goals. Less innovative firms tend to create environments of independence, whereas

innovative ones create environments of co-operation. Highly innovative companies also appear to place much more reasonable goal expectations, and try not to overload individuals with projects. The prevalent belief being that too many projects spread effort too thinly, leading individuals to step from the surface of one to the next. These conditions create time pressures, which militate strongly against innovativeness.

Continuity of Slack

Slack is the cushion of resources, which allows an organization to adapt to internal and external pressures. Slack has been correlated positively to innovation. Judge *et al.* (1997) note that it is not just the existence of slack but also the existence of slack over time that appears to have positive impact upon innovation. They find less innovative firms have slack but these firms appear to have experienced significant disruptions or discontinuities of slack in their past or were expecting disruptions in the future. Therefore innovativeness seems to be linked with both experience and expectations of slack resources. It can be hypothesized that slack, and future expectations of uninterrupted slack, provide scope for the organization and its members to take risks that they would not take under conditions of no slack, or interruptions in slack. Organizationally, this would appear to indicate the need for generating a base-line stock of slack in a variety of critical resources (such as time and seed funding for new projects). Figure 12.1 depicts the characteristics of an innovative/creative organization.

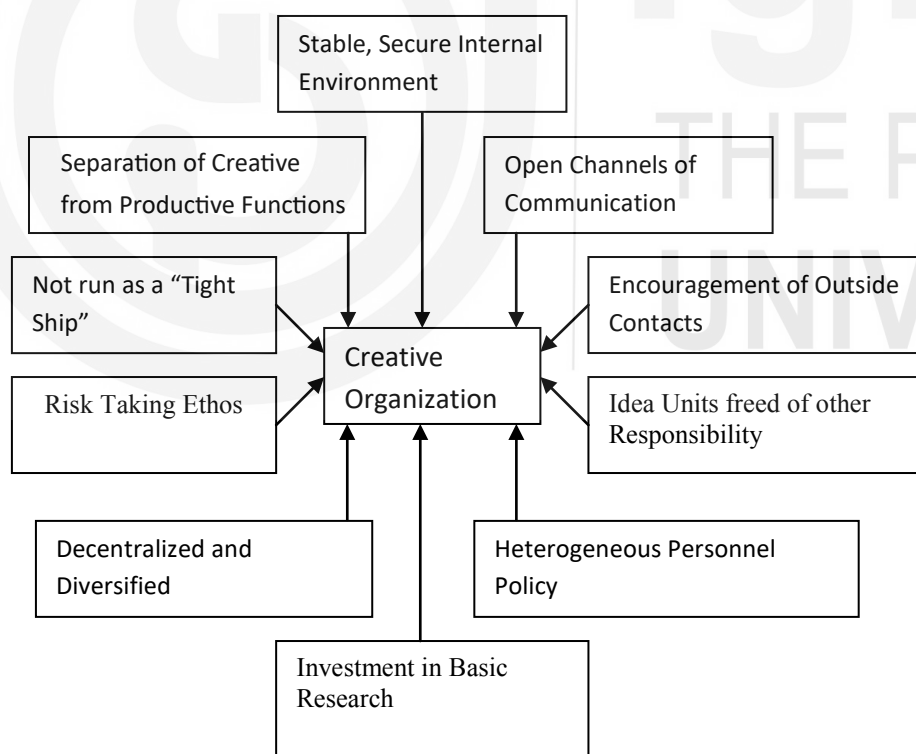


Figure 12.1: Characteristics of an Innovative/Creative Organization

12.5 THE INDIVIDUAL AND INNOVATION CULTURE

People play a role in organizational culture. Organizations need to consider the type of employees that can most effectively drive innovation. From a

Strategic Enablers

diverse range of research (psychology to management) it has been found that a core of reasonably stable personality traits characterizes creative individuals. A select few of these are listed below.

- High valuation of aesthetic qualities in experience
- Broad interests
- Attraction to complexity
- High energy
- Independence of judgement
- Intuition
- Self-confidence
- Ability to accommodate opposites
- Firm sense of self as creative
- Persistence
- Curiosity
- Energy
- Intellectual honesty
- Internal locus of control (reflective/introspective)

Although there appears to be general agreement that personality is related to creativity, attempts to try and use this inventory type of approach in an organizational setting as predictor of creative accomplishments is fraught with dangers, and is hardly likely to be any more useful than attempts at picking good leaders through the use of trait theory approaches. Nevertheless it does highlight the need to focus on individual actors, and to try and nurture such characteristics or at least bring them out, if necessary, in an organizational setting.

Cognitive Factors and Innovation

Cognitive factors also appear to be associated with the ability to innovate. Research appears to indicate that a number of cognitive factors are associated with creativity. For example, medical psychology indicates differences in cognitive processing, ascribing left cerebral cortex to rational thinking, and the right brain to intuition.

Cognitive parameters affecting idea production are given below:

- Associative fluency
- Fluency of expression
- Figural fluency
- Ideational fluency
- Speech fluency
- Word fluency
- Practical ideational fluency
- Originality

- Fluency
- Flexibility

Personal Motivational Factors Affecting Innovation

At the individual level numerous motivation-related factors have been identified as drivers of creative production. The key ones are presented below:

Intrinsic Versus Extrinsic Motivation

Intrinsic motivation is a key driver of creativity (Amabile, 1990; Baron and Harrington, 1981). In fact extrinsic interventions such as rewards and evaluations appear to adversely affect innovation motivation because they appear to redirect attention from “experimenting” to following rules or technicalities of performing a specific task. Furthermore, apprehension about evaluation appears to divert attention away from the innovation because individuals become reluctant to take risks since these risks may be negatively evaluated. On the contrary, in order to be creative, individuals need freedom to take risks, play with ideas and expand the range of considerations from which solutions may emerge.

Challenging Individuals

Open ended, non-structured tasks engender higher creativity than narrow jobs. This occurs by virtue of the fact that people respond positively when they are challenged and provided sufficient scope to generate novel solutions. It appears that it is not the individual who lacks creative potential but it is the organizational expectations that exert a primary debilitating effect upon the individual’s inclination to innovate (Shalley and Oldham, 1985).

Skills and Knowledge

Creativity is affected by relevant skills such as expertise, technical skills, talent, etc. However such domain-related skills can have both positive as well as negative consequences. Positively, knowledge enhances the possibility of creating new understanding. Negatively, high domain-relevant skills may narrow the search heuristics to learnt routines and thereby constrain fundamentally new perspectives. This can lead to functional “fixedness”.

At a more macro-level Schneider *et al.* (1996) suggest that organizations may attract and select persons with matching styles. Organizational culture, as well as other aspects of the organization, may be difficult to change because people who are attracted to the organization may be resistant to accepting new cognitive styles. When a change is forced, those persons attracted by the old organization may leave because they no longer match the newly accepted cognitive style. Among other things, this culture-cognitive style match suggests that organizational conditions (including training programmes) supportive of creativity will be effective only to the extent that the potential and current organizational members know of and prefer these conditions.

Box 12.2: List of Qualities that Describe Innovators

- **Challenges status quo:** Dissatisfied with current reality, questions authority and routine and confronts assumptions.
- **Curious:** Actively explores the environment, investigates new possibilities, and honours the sense of awe and wonder.
- **Self-motivated:** Responds to deep inner needs, proactively initiates new projects, intrinsically rewarded for efforts.
- **Visionary:** Highly imaginative, maintains a future orientation, thinks in mental pictures.
- **Entertains the fantastic:** Conjures outrageous scenarios, sees possibilities within the seemingly impossible, honours dreams and daydreams.
- **Takes risks:** Goes beyond the comfort zone, experimental and non-conforming, courageously willing to “fail”.
- **Peripatetic:** Changes work environments as needed; wanders, walks or travels to inspire fresh thinking; given to movement and interaction.
- **Playful/humorous:** Appreciates incongruities and surprise, able to appear foolish and child-like, laughs easily and often.
- **Self-accepting:** Withholds compulsive criticism of their own ideas, understands “perfection is the enemy of the good,” unattached to “looking good” in the eyes of others.
- **Flexible/adaptive:** Open to serendipity and change, able to adjust “game plan” as needed, entertains multiple ideas and solutions.
- **Makes new connections:** Sees relationships between seemingly disconnected elements, synthesizes odd combinations, distills unusual ideas down to their underlying principles.
- **Reflective:** Incubates on problems and challenges; seeks out states of immersion; ponders, muses and contemplates.
- **Recognizes (and re-cognizes) patterns:** Perceptive and discriminating, notices organizing principles and trends, sees (and challenges) the “Big Picture.”
- **Tolerates ambiguity:** Comfortable with chaos, able to entertain paradox, doesn’t settle for the first “right idea.”
- **Committed to learning:** Continually seeks knowledge, synthesizes new input quickly, balances information gathering and action.
- **Balances intuition and analysis:** Alternates between divergent and convergent thinking; entertains hunches before analyzing them; trusts their gut, uses their head.
- **Situationally collaborative:** Balances rugged individualism with political savvy, open to coaching and support, rallies organizational support as needed.
- **Formally articulate:** Communicates ideas effectively, translates abstract concepts into meaningful language, creates prototypes with ease.
- **Resilient:** Bounces back from disappointment, learns quickly from feedback, willing to “try, try again”.
- **Persevering:** Hardworking and persistent, champions new ideas with tenacity, committed to follow-through and bottom-line results.

Source: “Free the Genie” series, a set of 12 creative thinking booklets, by Mitchell Ditkoff, President, Idea Champions.

12.6 FOSTERING CREATIVITY AND THE CREATIVE PROCESS

Good problem solving occurs when managers have many viable, creative alternatives to consider. To inspire employees to approach problems creatively and to nurture a creative environment, organizations follow three general approaches. These include hiring creative individuals, applying specific creativity-enhancement techniques, and developing a creative organization.

Creative behaviour is defined as production ideas that are both new and useful. Creative ability is the ability to produce ideas that are both new and useful. These definitions may seem constraining, since the usefulness of some truly creative alternative might not be immediately evident. One scholar has addressed this dilemma by differentiating between originality and creativity. Both motivation and a proper setting may be necessary if innate creative ability is to blossom into creative output.

Creativity involves more than the sudden moment of inspiration in which an idea suddenly flashes in the brain. Instead, as shown in figure 12.2 below, there are four stages to the creative process: preparation, incubation, insight, and verification.



Figure 12.2: The Creative Process

Preparation involves gathering, sorting, and integrating information and other materials to provide a solid base for a later breakthrough. The discoveries of penicillin, the benzene ring, or gravity, while each involved a moment of insight, would have been impossible without a firm grasp of related information. During the incubation stage, the mind is not consciously focused on the problem. The individual may be relaxed, asleep, reflective, or otherwise involved. The insight (“Eureka!”) stage is the familiar, sudden moment of inspiration. While this is what we often think of as creativity, it is only one step in the creative process.

Finally, verification is necessary. Here, the individual carries out the chores involved in carefully checking facts to support the insight, carrying out research to determine that the DNA molecule is in fact a double helix or that a meteorite did really create a dust cloud that led to the extinction of the dinosaurs. This process further supports the contention that creativity does not just happen. It is a thorough and often-painstaking activity.

Activity 1

- a) Explain giving illustration, how creativity and innovation can play an important role in the success of an organization.

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- b) Give the name of the company under the given industry heads, which have adopted innovation strategy and have succeeded to a large extent.

Industry	Company
Petroleum	
Textile	
Pharmaceutical	
Telecom	

12.7 TECHNIQUES FOR ENHANCING CREATIVITY

A wide variety of popular techniques have been developed to enhance creativity, as described in this section.

Gordon Technique

William J.J. Gordon worked with creative-thinking groups and had a creative variety of other pursuits. He was concerned that people, when asked to come up with a creative new idea, would instead incrementalize. That is, they would take an available alternative and improve it bit by bit. While this might lead to marginally better alternatives the alternatives probably would not be real breakthroughs. Gordon decided that one way to avoid this problem would be simply not to tell people what they were inventing. Thus, the Gordon technique uses an initial focus on function. Rather than being told to build a better mousetrap, the group might first be told that the focus was capturing. Instead of the group being instructed to design an improved knife, the function could be given as severing.

Synectics

Gordon also developed a well-known technique called synectics. Synectics means “the joining of apparently unrelated elements.” First, very different sorts of people are put together in synectics groups in order to get a real diversity of perspectives. Second, synectics relies heavily on the use of analogies. Synectics techniques have been widely adopted by both businesses and educational institutions. Three synectics tools according to Gordon are direct analogy, personal analogy, and fantasy analogy.

Direct Analogy: This involves looking for parallel facts, knowledge, or technology in a different domain from the one being worked on. For instance, can we think of anything similar that occurs in nature?

Personal Analogy: With this approach, synectics group members try to identify psychologically with key parts of the problem. In one case, for example, the group was asked to design a mechanism that would run a shaft turning at 400 to 4000 rpm so that the power-takeoff end of the shaft would turn at a constant 400rpm. To address this question, members of the group metaphorically entered the box and tried to use their bodies to attain the required speed without undue friction.

Fantasy Analogy: Sigmund Freud saw creativity as the fulfillment of a wish or fantasy. Fantasy analogy asks how in many wildest dreams can I make this happen? Gordon gives the example of a synectics group with the task of inventing a vapour-proof closure for space suits. Their solution was a spring mechanism based on the fantasy analogy of rows of trained insects clasping claws to hold shut the closure.

There is more to synectics than just the use of analogy. The technique follows a structure problem-solving sequence in which a client and other participants interact to develop a workable solution to the client's problem. For instance, after the problem has been introduced and discussed, there is a "Springboards" stage in which the problem is opened up by asking the client to convert concerns, opinions, and desires into statements such as "I wish..." or "How to ...". Later, after an initial idea has been developed and refined, an "itemized response" stage requires the client to think of three useful aspects or advantages of the idea and to generate key concerns. Still later, after the group works to modify the suggestion to overcome these concerns, the "possible solution" is checked for elements of newness and feasibility and whether there is sufficient commitment to the solution to take additional steps. Finally, the client lists actions to be taken to implement the solution, including timing and the personnel to be used.

Betsy Means, vice president and director product management for the values product group of Citibank regularly uses analogies to name a new product. She first brainstorms with a group to come up with a new fraction of an idea and then builds on it by bringing together ideas from unrelated disciplines. Through her efforts, Means in a single year signed up more than 2 million new customers for Citibank's Visa and MasterCard.

Idea Checklists

Several idea checklists have been developed to enhance creativity. These involve asking a series of questions about how we might use something that we already have. For example, one checklist of idea-spurring questions is called SCAMPER (Substitute? Combine? Adapt? Modify or magnify? Put to other uses? Eliminate or reduce? Reverse or rearrange?). Here's an example of adapting: Clarence Birdseye worked as a fur trader in Labrador before World War I. He noted that Inuit preserved fish by quick-freezing and that the fish, when thawed, were flaky and moist. Birdseye adapted this process to make quick-frozen food available to the general public. This replaced the old slow freeze process that left food dry and tasteless. The huge success of quick frozen food led to the creation of General Foods.

Kiichiro Toyoda, the founder of Toyota, sought ways to eliminate large inventories and the need for warehouses. American supermarkets fascinated him, and he noted that they require vast amounts of food that can't be stored on site because of spoilage and space considerations. When supplies run low, the staff contacts the appropriate supplier and items arrive "just in time." Toyota adopted this concept and streamlined its operation, eliminating waste and warehouses and reducing costs dramatically. Toyota's "just-in-time" approach gave it a huge competitive edge. Just in time is now being adopted worldwide.

George Washington Carver asked the question “How can peanuts be put to other uses? And came up with over 300 applications. Many creative ideas have resulted from asking how waste products could be put to other uses. Rubber bands are made from surgical tubing; garbage is compressed into construction blocks; petrochemical waste is sold as silly putty. The Goodyear Tire Company has a pollution-free heating plant in Michigan that uses discarded tires as its only fuel.

Perhaps the best-known listing technique is the “73 idea-spurring questions” devised by Osborn. This checklist can be applied to any alternative. Here are some of the questions.

- Put to other uses? New ways to use as is? Other uses if modified?
- Adapt? What else is like it? What other ideas does this suggest? Does past offer parallel? What could I copy? Whom could I emulate?
- Minify? What to subtract? Smaller? Condensed? Miniature? Lower? Shorter? Lighter? Omit? Streamline? Split up? Understate?
- Substitute? Who else instead? What else instead? Other ingredient? Other material? Other process? Other power? Other place? Other approach? Other tone of voice?
- Rearrange? Interchange components? Other pattern? Other layout? Other sequence? Transpose cause and effect? Change pace? Change schedule?
- Combine? How about a blend, an alloy, an assortment, an ensemble? Combine units? Combine purposes? Combine appeals? Combine ideas?

Attribute Listing

According to the developer of attribute listing, Robert Crawford, “Each time we take a step we do it by changing an attribute or a quality of something, or else by applying that same quality or attribute to some other things. There are two forms of attribute listing: attribute modifying and attribute transferring.

With attribute modifying, the main attribute of the problem object is listed. Then ways to improve each attribute are listed. For instance, the technique might be used to concentrate on ways to improve the running shoe attributes of weight, stability, cushioning, and durability. Attribute transferring is similar to direct analogy in synectics. Attributes from one thing are transferred to another.

Checkerboard Method

The checkerboard method, also called morphological analysis, is an extension of attribute modifying. Specific ideas for one attribute or problem dimension are listed along one axis of a matrix. Ideas for a second attribute are listed along the other axis. If desired, a third axis (and attribute) can be added. Figure 12.3 shows the checkerboard.

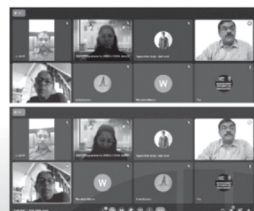
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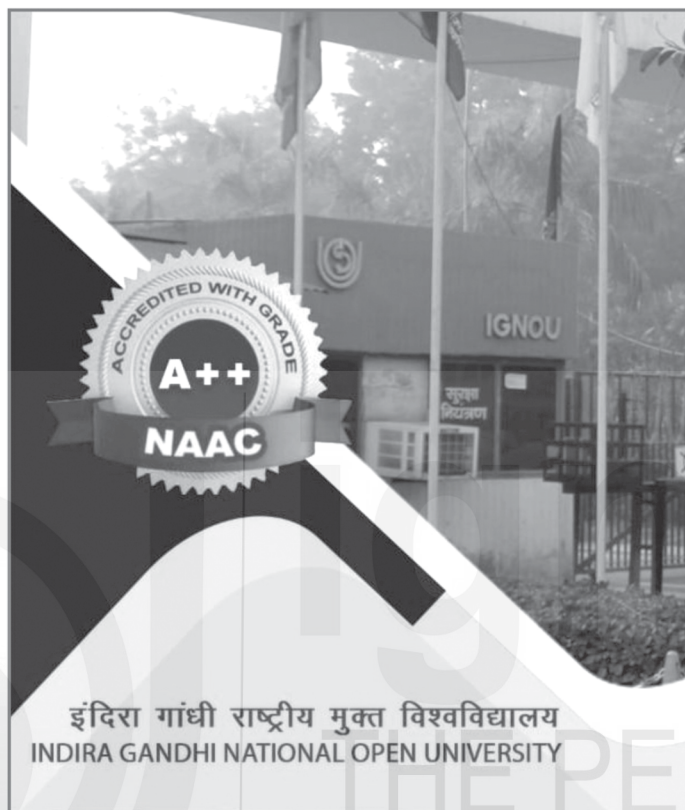
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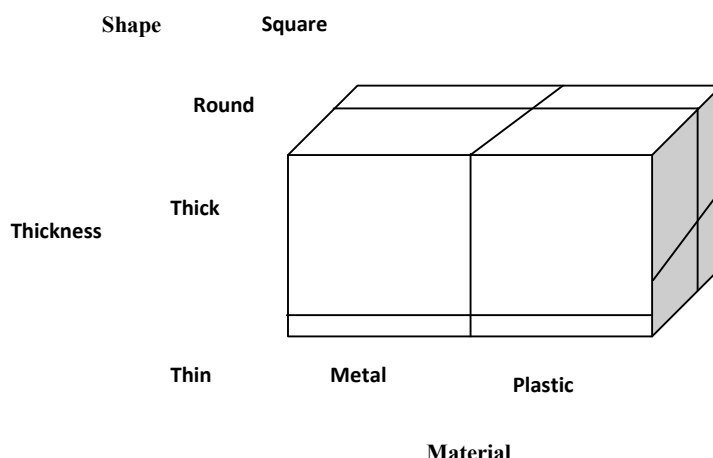


Figure 12.3: The Checkerboard

The cells of the matrix then provide idea combinations. For instance, the axes for a vehicle might be type of energy source (e.g., rollers, air, water, rails), and type of vehicle (e.g. cart, chair, sling, bed). The above figure shows a simple application of the checkerboard method to the design of paper clips. The benefit of the checkerboard method of analysis is that it makes us aware of all possible combinations of the attributes. Many, of course, will prove to be little value, but others may be worthwhile. Like other creativity enhancement techniques, the checkerboard method makes us view the world from a different perspective. It is very useful for producing large numbers of new ideas.

Retroduction

We are the slaves of our assumptions; they dictate the way we behave. Retroduction involves changing an assumption. This may serve two purposes. First, our assumptions may be wrong. Second, even if our assumptions are correct we may gain valuable new perspectives from looking at things from a different angle. Albert Einstein, for instance, revised Isaac Newton's assumption that space is flat to the assumption that space is curved and developed a new perspective on time and space.

As a simple example of the power of assumptions, consider paper clips, the subject. The standard Gem paper clip, invented in 1899, accounts for most of the 20 billion paper clips sold every year. More than 100 alternative designs have patented, varying in size, material, and shape. "Ring" clips, "owl" clips, "arrowhead" clips, "butterfly" clips, and many others have been offered, and their inventors present compelling cases for their superiority. Nevertheless, they haven't made a noticeable dent in Gem's market superiority. The reason is that inventors share a common and incorrect assumption, that paper clips are used to clip sheets of paper. In fact, research shows that only 20 percent of paper clips are used to hold papers. The rest are twisted or broken by people during phone conversations, unwound to clean pipes, nails, or ears, used to reinforce eyeglasses, or put to other creative uses. The Gem, unlike its competitors, can easily be taken apart and reshaped.

One retroduction technique says, "Suppose X were Y." For instance, "suppose custodians were chief executives." Another technique pairs apparently distinct concepts, such as power and satisfaction or perception and structure,

and sees what new alternatives might be suggested. Yet another asks “What if?” for example, what if employees could design their own jobs? What if we viewed customers as owners of the firm? One individual who applied these retroduction techniques generated such questions as “what are the structural irregularities of semiconductors?” and “can arteries have rashes?” Each of these questions is now the subject of study and debate, the first among physicists and the second among researchers and disease processes. Henry Ford questioned the practice of moving workers to material asking “What if we moved the work to the people?” This questioning led to the birth of the assembly line. Retroduction offers new perspectives and helps free people from mental ruts.

Here’s a final example: For years, bankers assumed that customers preferred human tellers. In the early 1980s Citibank felt that installing automotive tellers would help it cut costs. However, since Citibank executives assumed people would prefer not to use machines, they reserved human tellers for people with large accounts and relegated smaller depositors to the machines. The machines proved unpopular and Citibank stopped using them, taking the failure as proof that its assumption was correct. Later, another banker challenged this assumption. He asked, in effect, “What if people really like to use automatic teller? What if the Citibank customers who used the machines simply resented being treated as second class citizens?” He brought back the automatic tellers with no “class distinction” and they were an immediate success.

12.8 BUILDING CREATIVE ORGANIZATIONS

Venture Teams

A venture team is a temporary grouping of organization members for generating new ideas. So that creative thinking is not stifled, team members are freed of the organization’s bureaucracy and in many cases have a separate location and facilities. These separate entities are known as skunk works. Major corporations such as IBM, 3M Company, Dow Chemical, and Texas Instruments have used venture teams to solve technical problems and promote change. Ford Motor Company used a skunk works to keep the new mustang alive. Faced with tight budgets, tough time constraints, and an uncertain vision of the new mustang, Ford formed the 400 members “Team Mustang” Team members thought of themselves as independent stockholders of the “Mustang Car Company” in converted furniture warehouse, got approval to move Ford into “Chunk teams” with responsibility for every “chunk” of the car. Mustang Car Company did away with many elements of the traditional hierarchy and many restrictive rules and procedures. The result: the fundamentally redesigned new Mustang was completed in three years and for about \$700 million-25 percent less time and 30 percent less money than for any comparable new car programme in Ford’s recent history.

Idea Champions

An idea champion is a member of the organization who is assigned responsibility for the successful implementation of a change. The idea champion may be a senior manager or a non-manager, such as the inventor of the idea that has prompted the change. An idea champion will fight resistance to change and will actively pursue resources necessary to carry

out the change. Idea champions may be critical to the success or failure of change. For example, Texas Instruments reviewed 50 successful and 50 unsuccessful technical projects. One consistent finding was that every failure also lacked an idea champion. As a result, Texas Instruments set up as its number one criterion for project approval the presence of an idea champion.

Intrapreneurship

Many people have praised the flexibility, creativity, risk taking, and energy that are often associated with small firms and intrapreneurship and have asked how these elements might be instilled in larger organizations. Intrapreneurship is the name given to intrapreneurial activities within a larger organization, and intrapreneurs are essentially internal entrepreneurs.

Intrapreneurs and entrepreneurs have many things in common. For instance, they value creativity and autonomy and have strong desire to achieve. On the other hand, since intrapreneurs work within a corporate system, they face the benefits and constraints of that system. Unlike entrepreneurs, they operate under a corporate accounting system and must report to hierarchical superiors. They do not personally face the financial risks that entrepreneurs do, nor do they enjoy the same rewards. They can draw on the rich financial resources of the corporation.

Intrapreneurs may need different competencies to succeed than do entrepreneurs. For instance, intrapreneurs must be somewhat skilled at organizational politics, something that entrepreneurs may find reprehensible and that may, in fact, motivate them to work for themselves. Further, while entrepreneurs must provide their own goals and rewards, intrapreneurs are within the reward system of the formal organization.

Two major factors foster intrapreneurial success. **First**, sponsorship of intrapreneurs is important. Sponsors ensure that the intraprise gets the required resources, and they can help temper the grievances of those who feel threatened by the innovation. Many intrapreneurs have several sponsors. Lower-level sponsors to fend off threatening strategic attacks. **Second**, there must be suitable rewards for intrapreneurship. Traditional rewards for success do not match the risks of innovating or intrapreneuring. Also, the basic reward in most companies is promotion, which does not work well for most intrapreneurs; they seek freedom to use their intuition, take risks, and invest the company's money in building new businesses and launching new products and services. For this reason, a key reward for intrapreneurs is intra-capital. Intra-capital is a discretionary budget earned by the intrapreneur and used to fund the creation of new intra prizes and innovation for the corporation.

Hoping to grow and compete in a fast paced market, Bell Atlantic turned to intrapreneurship, with great success. Within a few years, more than 130 intrapreneurs had championed more than 100 projects, at least 15 products were on or near the market, and 15 patents had been awarded. Potential revenues estimated from the projects total a minimum of \$100 million within five years. Similarly, at Xerox, many creative ideas were lost before being turned into marketable products. As a result Xerox recognized the need to merge intrapreneurs within the corporation. The company formed

Xerox, Technology Ventures (XTV), and a venture capital group that allows Xerox to bring creative products to the market through intrapreneurships. XTV has become so successful that it is now a role model for other firms.

Creativity and Diversity

People referring in gender, age, race disability status and sexual orientation bring to organizations a variety of attitudes, values, and perspectives as well as a broad and rich base of experience to address a problem. As a result as the group became more diverse, the potential for creativity is enhanced. Innovative organizations have generally done a better job than others in eradicating racism and sexism, and they tend to employ more women and nonwhite men than do less innovative firms. In addition, brainstorming groups made up of diverse ethnic and racial groups produce higher-quality ideas than do homogeneous groups. Further, the presence in groups of individuals holding minority views lead to critical analysis of decision issues and alternatives, resulting in consideration of a larger number of alternatives and more thorough examination of underlying assumptions. And because homogeneous groups tend to value conformity and agreement and their members are sometimes afraid to “rock the boat,” such groups often discourage critical thinking. Because of this, diversity may foster more open, honest, and effective decision-making. Taken together, this all suggests that diversity can yield many benefits for decision-making creativity. However, diversity may also increase the potential for misunderstandings and increase conflict and anxiety among members. The challenge is to manage cultural diversity in such a way as to capture its benefits while minimizing potential problems.

12.9 COMPANY PROGRAMMES TO ENHANCE CREATIVITY

Firms are using special programmes to foster their employees' creativity. Many send their employees on retreats and outings to jolt them out of routine ways of thinking. Quaker oats Co.'s executives go horseback riding when they need fresh approaches to budget and marketing problems. American Greeting Co.'s licensing unit, the characters from Cleveland, which created Strawberry shortcake and the care Bears, go for half a dozen weekend retreats in the woods each year, where their creative personnel brainstorm, play games, and sketch to come up with creative ideas.

At Omron Corp., a maker of electronic controls, midlevel employees attend a monthly juku, or cram school, where they try to think plan as if they were 19th century warlords, private detective, or formula one race car drivers. Fuji film asks its senior managers to study topics such as the history of Venice and the sociology of apes. While such exercise may sometimes appear bizarre or even humorous, they encourage the employees to break out of their corporate shells and think in different ways.

Management at the Polaroid Corporation faced situation in which it needed to find a way to develop new photographic products that would enable the company to grow and regain profitability.

- a) Read the background information on Polaroid in the real world management challenge and then indicate what would be done to handle situation if you were the CEO of Polaroid later you can compare your recommendations with what management actually did in that situation.

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- b) Think of any two Indian companies which have developed special programmes to enhance the creativity of their employees.

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12.10 SUMMARY

The success and survival of companies in today's marketplace depend on creativity, innovation, discovery and inventiveness. The challenge, therefore, for many companies is to be innovative and creative to bring to the market a stream of new and improved, added value, products and services that enable the business to achieve higher margins and thus profits to re-invest in the business. Innovation can be defined as the successful exploitation of new ideas.

Virtually all companies talk about innovation and many may actually attempt to "do it", but only a few actually succeed in doing it. To be creative, an organization requires an innovation strategy, a strategy that promotes the development and implementation of new products and services. The origin of creativity and innovation lies in a shared vision and mission, which are focused on the future. Furthermore, the vision and mission of a creative and innovative organization are also customer and market oriented-focusing on solving customers' problems among other things. In addition, a structure that supports creativity and innovation, values like flexibility, freedom and cooperative teamwork is required to promote creativity and innovation. Support mechanisms should be present in an organization to create an environment that will promote creativity and innovation. Rewards and recognition and the availability of resources, namely time, information technology and creative people, are mechanisms that play this role.

The key distinguishing factor between innovative and less innovative firms is the ability of management to create a sense of community in the workplace. Highly innovative companies behave as focused communities whereas less innovative companies units behave more like traditional

bureaucratic departments. The concept of autonomy appears to be one of central importance for building creative organizations. Operational autonomy encourages a sense of the individual and promotes entrepreneurial spirit, whereas strategic autonomy is more to do with the level of alignment with organizational goals. It appears that firms that are most innovative emphasize operational autonomy but retain strategic autonomy for top management.

Innovative companies appear to rely heavily on personalized intrinsic awards, both for individuals as well as groups. Less innovative companies tend to place almost exclusive emphasis on extrinsic awards. Extrinsic rewards are things such as pay increases, bonuses and shares and stock options. Intrinsic rewards are those that are based on internal feelings of accomplishment by the recipient. Highly innovative companies appear to place equal emphasis on the technical side as well as the social side of the organization. In other words, they look to nurture not only technical abilities and expertise but also promote a sense of sharing and togetherness. Slack is the cushion of resources, which allows an organization to adapt to internal and external pressures. Slack has been correlated positively to innovation. Moreover, it is not just the existence of slack but also the existence of slack over time that appears to have positive impact upon innovation.

Creativity involves more than the sudden moment of inspiration in which an idea suddenly flashes in the brain. There are four stages to the creative process: preparation, incubation, insight, and verification. Good problem solving occurs when managers have many viable, creative alternatives to consider. To inspire employees to approach problems creatively and to nurture a creative environment, organizations follow three general approaches. These include hiring creative individuals, applying specific creativity-enhancement techniques, and developing a creative organization. A wide variety of popular techniques have been developed to enhance creativity, which include Gordon Technique, Syntectics, Idea Checklists, Attribute Listing, Checkerboard Method and Retroduction.

12.11 KEY WORDS

Attribute Listing: In this technique, each time we take a step we do it by changing an attribute or a quality of something, or else by applying that same quality or attribute to some other things.

Checkerboard Method: The checkerboard method, also called morphological analysis, is an extension of attribute modifying. Specific ideas for one attribute or problem dimension are listed along one axis of a matrix. Ideas for a second attribute are listed along the other axis.

Gordon Technique: Gordon technique is a popular technique developed to enhance creativity. It uses an initial focus on function. For instance, instead of the group being instructed to design an improved knife, the function could be given as severing.

Idea Champions: An idea champion is a member of the organization who is assigned responsibility for the successful implementation of a change.

Idea Checklists: Several idea checklists have been developed to enhance creativity. These involve asking a series of questions about how we might use something that we already have.

Innovation: The intentional introduction and application within a role, group or organization of ideas, processes, products or procedures, new to the relevant unit of adoption, designed to significantly benefit the individual, the group, organization or wider society. Innovation and Creativity are often used interchangeably in the literature.

Intrapreneurship: Intrapreneurship is the name given to intrapreneurial activities within a larger organization, and intrapreneurs are essentially internal entrepreneurs.

Retrodution: People are the slaves of their assumptions; they dictate the way they behave. Retrodution involves changing an assumption.

Synectics: Gordon also developed a well-known technique called Synectics. Synectics means “the joining of apparently unrelated elements.”

Venture Teams: A venture team is a temporary grouping of organization members for generating new ideas.

12.12 SELF-ASSESSMENT QUESTIONS

- 1) Explain the concept of creativity/innovation? How does creativity contribute to the success of an organization? Give an example of a creative organization in the Indian context.
- 2) What are the various characteristics, which distinguish a creative organization from the others? What are the characteristics of creative individuals and how can a company tap their potential?
- 3) Discuss the various steps involved in creative process. How can creativity be encouraged within an organization?
- 4) Explain the various techniques available to foster creativity.

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